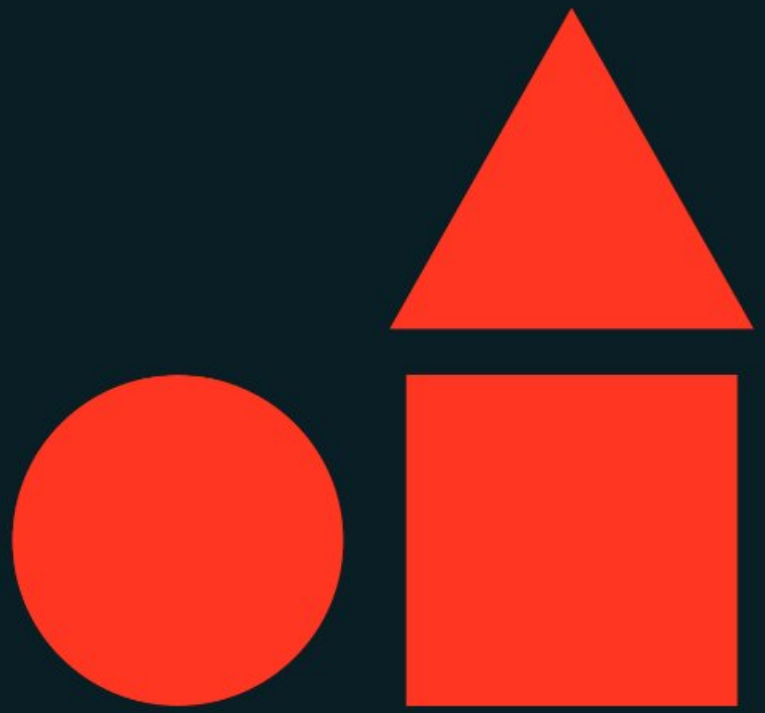
databricks

Deploy Workloads with Lakeflow Jobs



Databricks Academy

September 2025

Agenda

Course Sections
• Introduction to Data Engineering in Databricks
• Lakeflow Jobs Core Concepts
• Creating and Scheduling Jobs
• Advanced Lakeflow Jobs Features



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Course Learning Objectives

- Understand the role of Lakeflow Jobs within the Databricks ecosystem as a unified orchestration platform for data, analytics, and AI workloads.
- Design and implement data workloads using Directed Acyclic Graphs (DAGs), demonstrating the relationship between jobs, tasks, and dependencies.
- Configure various job scheduling options including manual, scheduled, file arrival, and continuous triggers to automate workflow execution.
- Implement advanced workflow features such as conditional task execution, run-if dependencies, for each loop and repair runs to create robust, fault-tolerant data pipelines.
- Apply best practices for production workloads including selecting appropriate compute options, implementing modular orchestration, and utilizing proper error handling techniques.



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Course Prerequisites (REQUIRED)



Fundamental Knowledge of the Databricks Platform

- Databricks Workspaces
- Apache Spark
- Delta Lake and the Medallion Architecture
- Unity Catalog



Intermediate Python Knowledge

- Creating functions and working with variables
- Reading data into DataFrames
- Applying basic transformations (select, filter, groupBy, etc.)



Experience Transforming Data with SQL

- Writing intermediate-level queries
- Creating and manipulating columns
- Aggregating data
- Filtering results



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Lab Exercise Environment



Technical Details

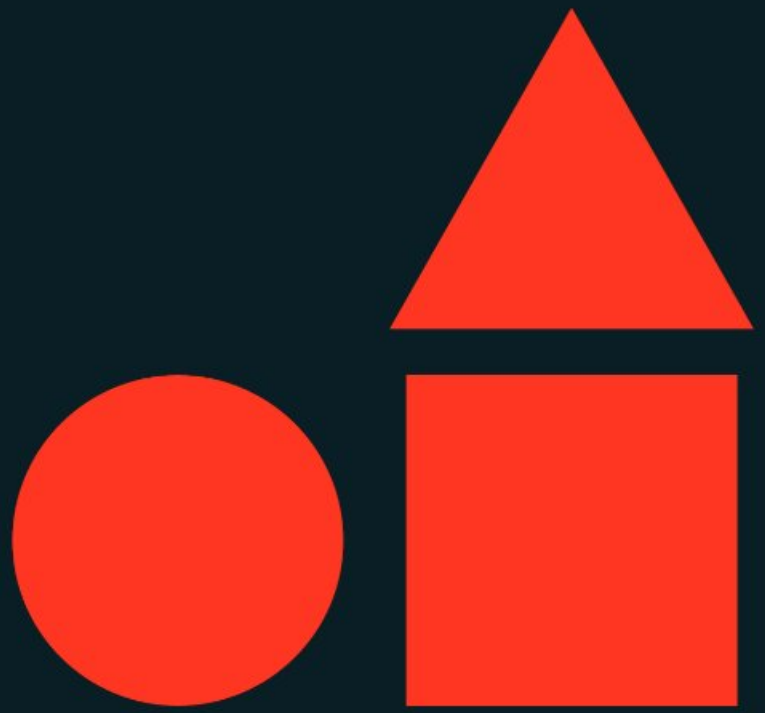
- Your lab environment is provided by Vocareum.
- It will open in a new tab.
- It has been configured with the permissions and resources required to accomplish the tasks outlined in the lab exercise.
- Third party cookies must be enabled in your browser for Vocareum user experience to work properly.
- Make sure to enable pop ups!



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Introduction to Data Engineering in Databricks



Deploy Workloads with Lakeflow Jobs

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Section Learning Objectives

- Understand the key components of data engineering on the Databricks platform, including Connect, Spark Declarative Pipelines and Jobs.
- Explain what Lakeflow Jobs are and their main benefits for unified orchestration of data, analytics, and AI workloads.
- Identify the core capabilities and use cases enabled by Lakeflow Jobs within the Databricks ecosystem.



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Agenda

Introduction to Data Engineering in Databricks
• Data Engineering in Databricks
• What is Lakeflow Jobs?



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Introduction to Data Engineering in Databricks

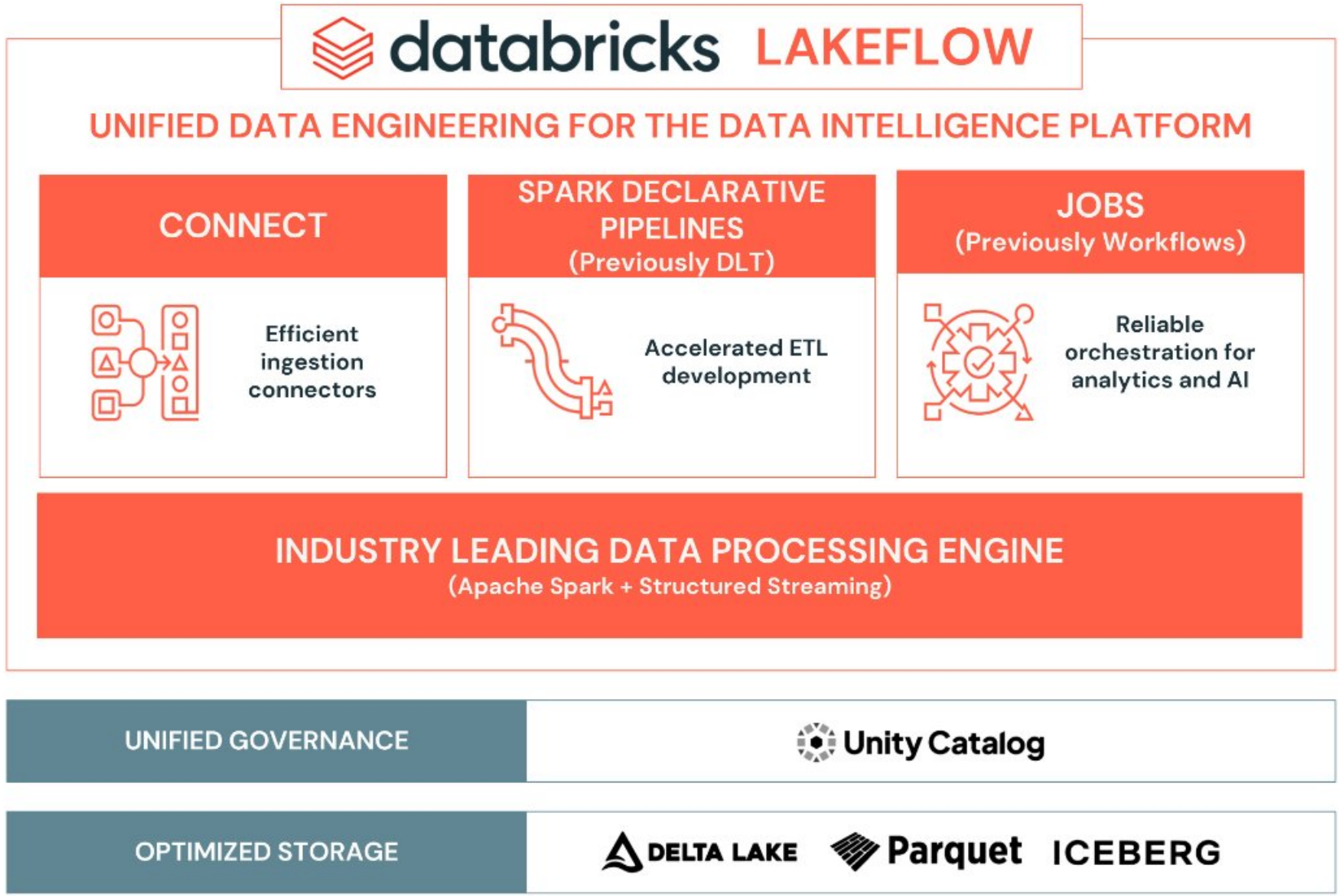
LECTURE

Data Engineering in Databricks



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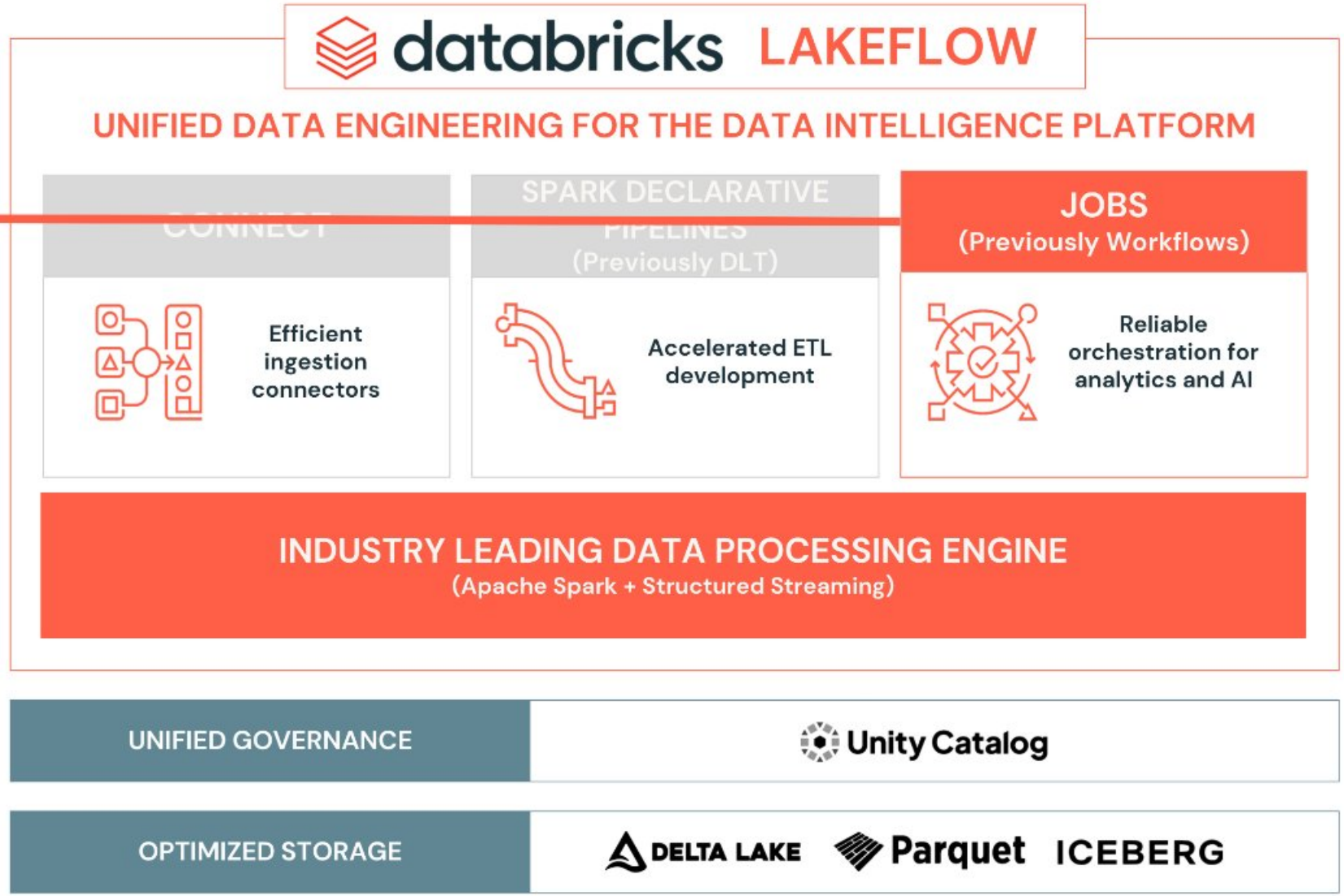
Data Engineering in Databricks



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Data Engineering in Databricks

This course will focus on **orchestrating every workload** (Notebooks, SQL, dashboards, Pipelines, and more)



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Introduction to Data Engineering in Databricks

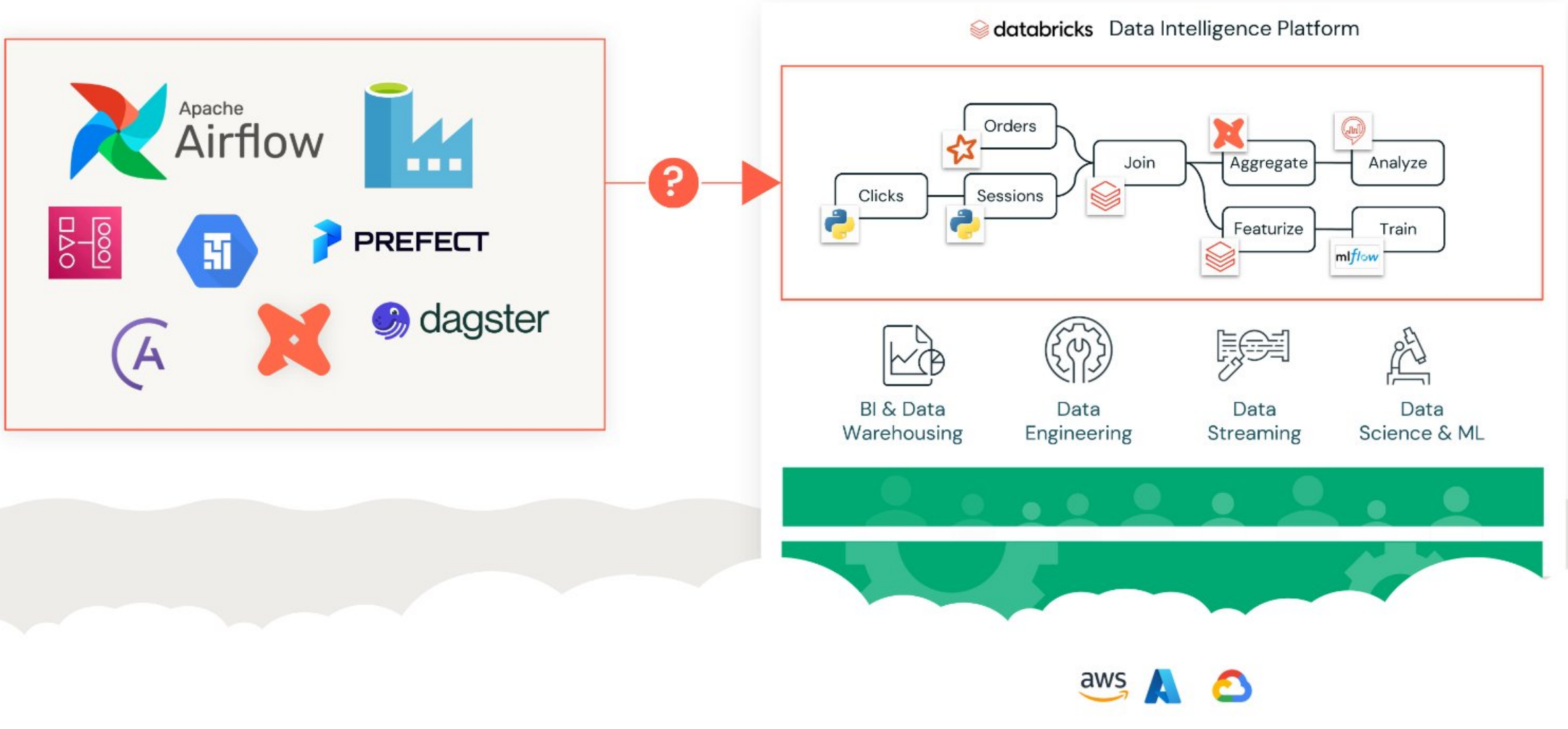
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What is Lakeflow Jobs?



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There are many ways to orchestrate your workloads



External orchestrators create challenges

Hard to use for many practitioners	Difficult to understand root cause when issues occur	Complex architecture to manage and maintain
Data teams are less productive	Bad data lowers value of downstream applications	Higher cost of ownership and lower reliability





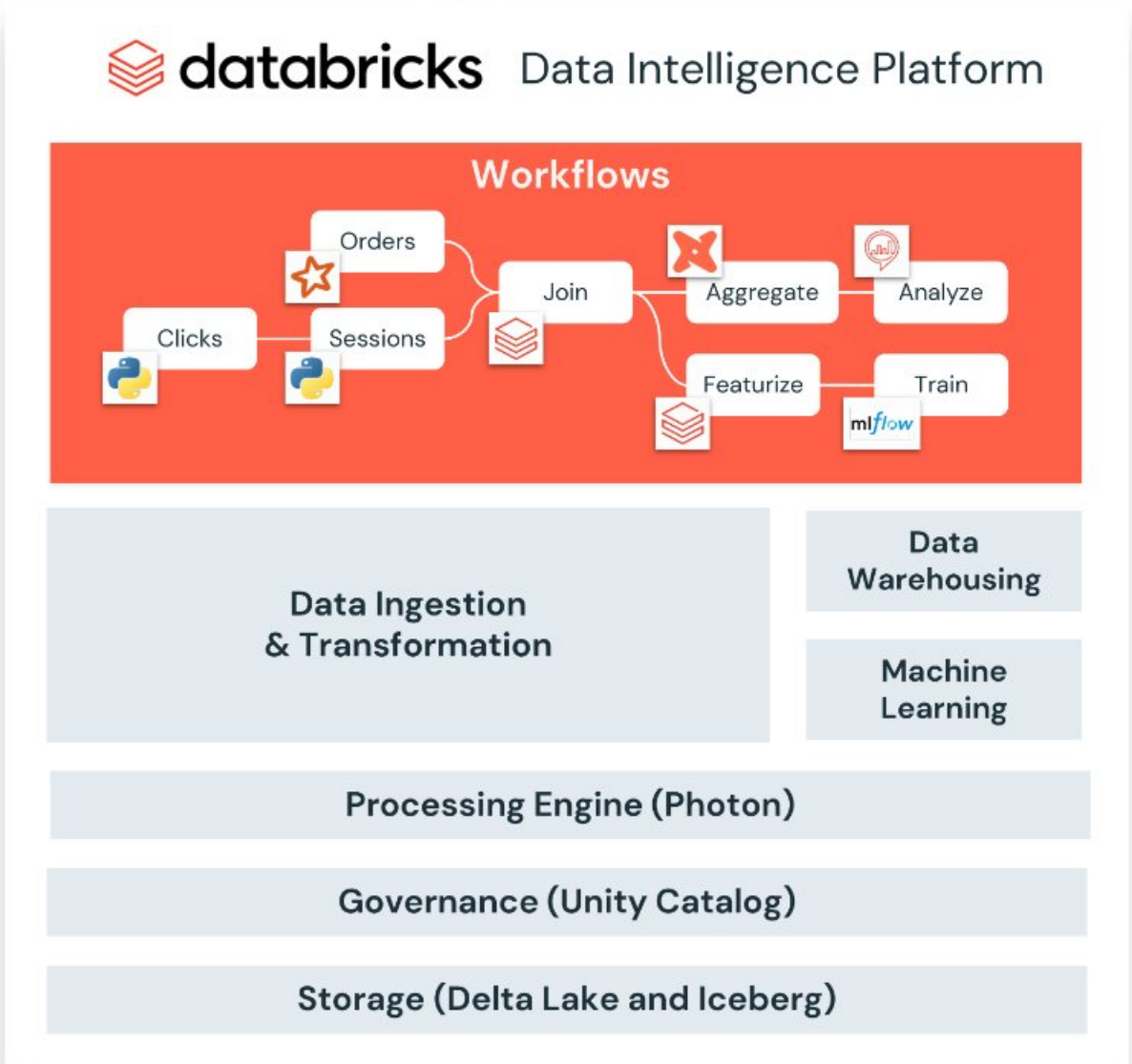
These tools are not unified with your Lakehouse





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What is Lakeflow Jobs?

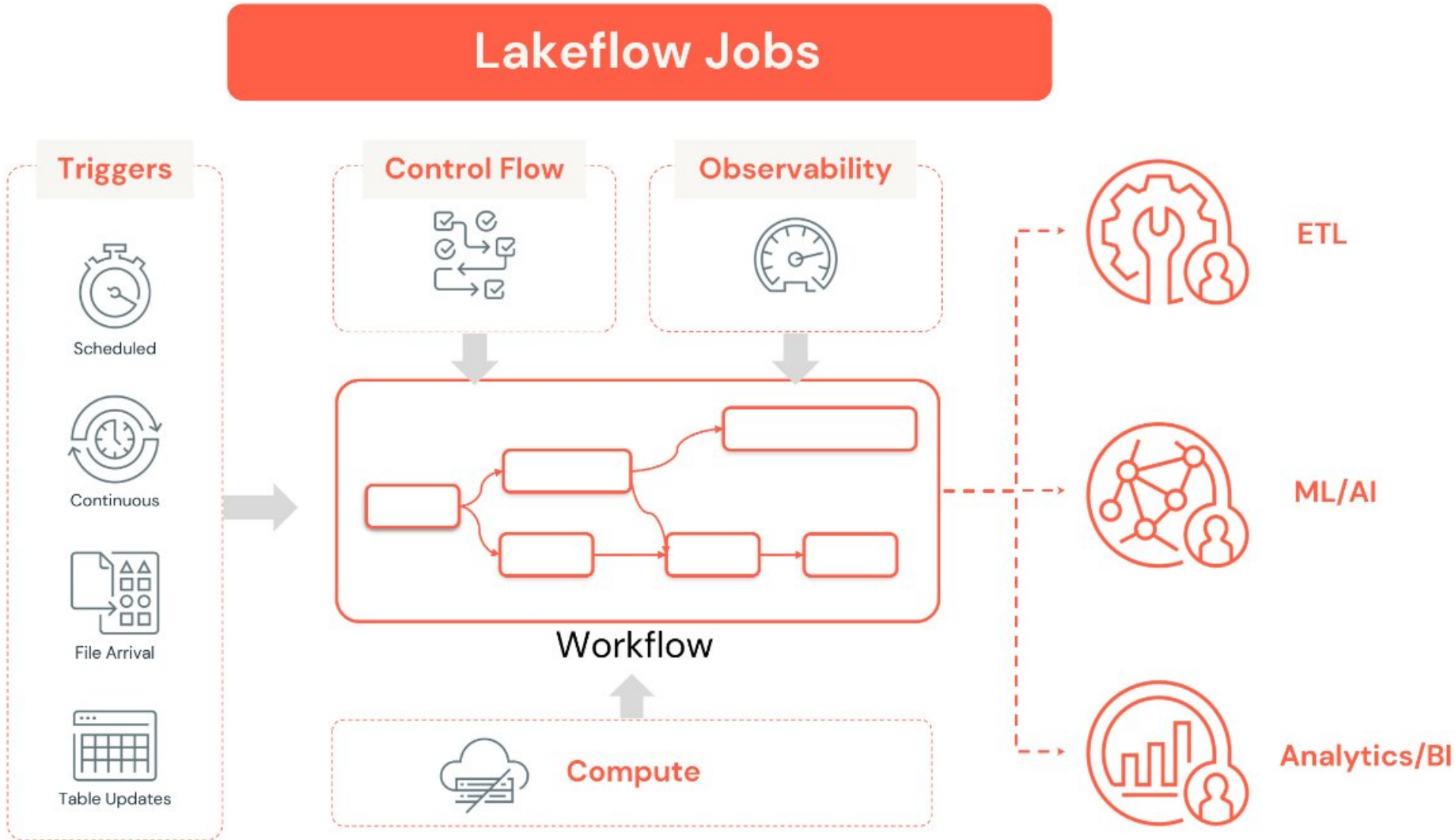


Jobs

Unified orchestration for data, analytics, and AI on the Data Intelligence Platform

- Simple authoring
- Actionable insights
- Proven reliability

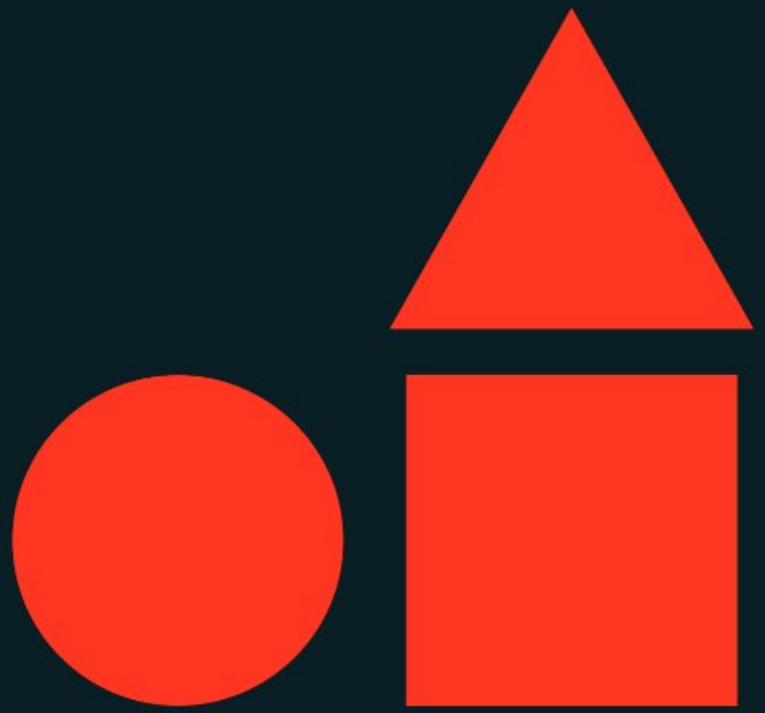
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Lakeflow Jobs Core Components



Deploy workload with lakeflow jobs

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Section Learning Objectives

- Understand the building blocks of Lakeflow Jobs, including the concepts of jobs and tasks.
- Identify and describe the various task types and configuration options available in Lakeflow Jobs.
- Explain how Directed Acyclic Graphs (DAGs) enable task orchestration and workflow patterns.
- Learn how to create and configure jobs using the Lakeflow Jobs UI.



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Agenda

Section Overview - Lakeflow Jobs Core Components
• Building Blocks of Lakeflow Jobs
• Task Orchestration
• Creating a Job Using the Lakeflow Jobs UI



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Lakeflow Jobs Core Components

LECTURE

Building Blocks of Lakeflow Jobs



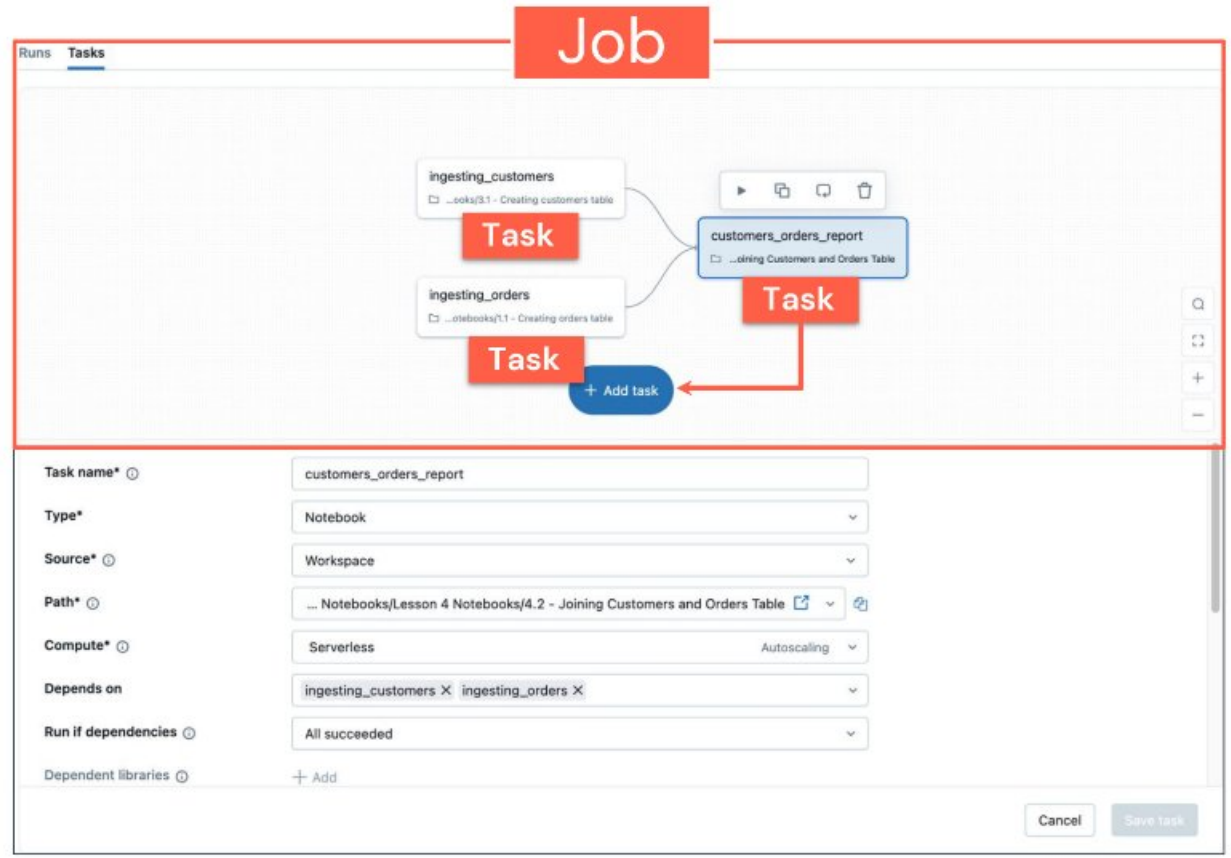
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Building Blocks of Lakeflow Jobs

Job and Tasks

- A **Job** is the primary resource for:
  - Scheduling
  - Coordinating
  - Running operations such as data processing, ETL, analytics, and machine learning workloads within the Databricks environment.

- A **task** is a **single unit of work** within a job that executes a specific workload such as a notebook, script, query and more.
- Each **job** consists of **one or more tasks**, which are the individual units of work that make up the job

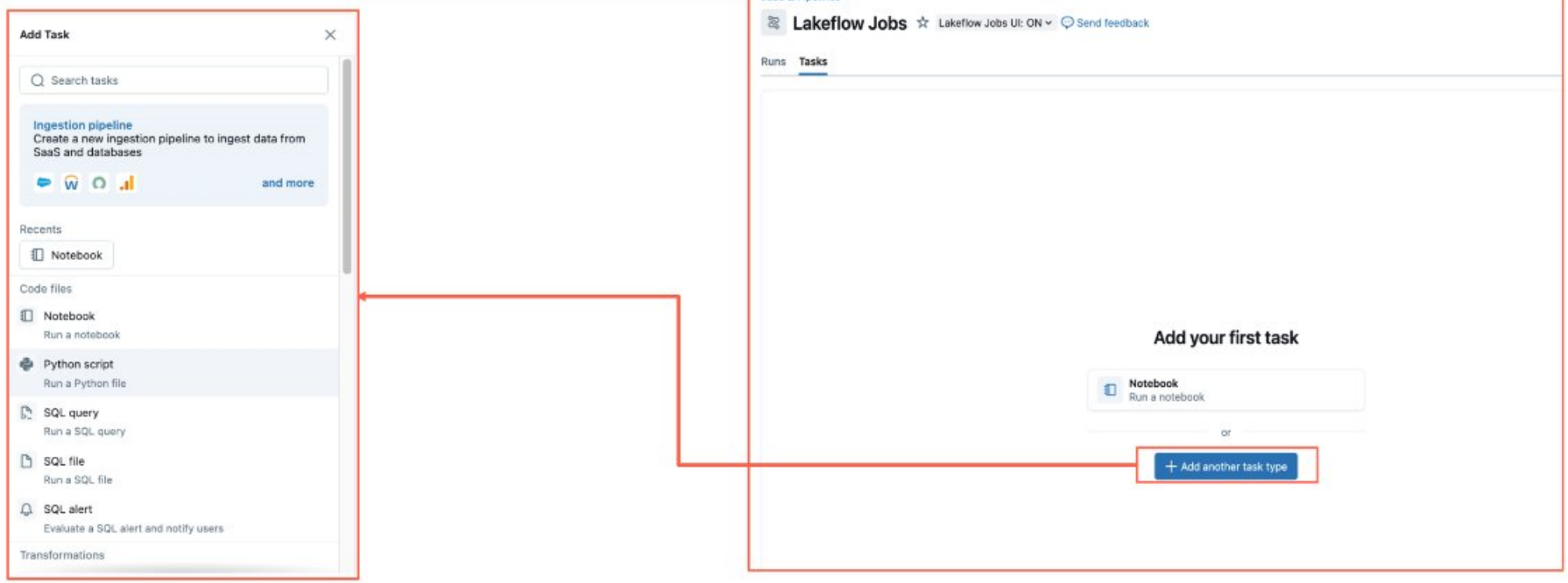


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Building Blocks of Lakeflow Jobs

Jobs consist of one or more **Tasks**

Databricks Notebooks | Spark Declarative Pipelines | SQL Files/ Queries | Python Scripts | Python Wheels | AI/BI Dashboards | Spark Submit | dbt | Java JAR file | Power BI

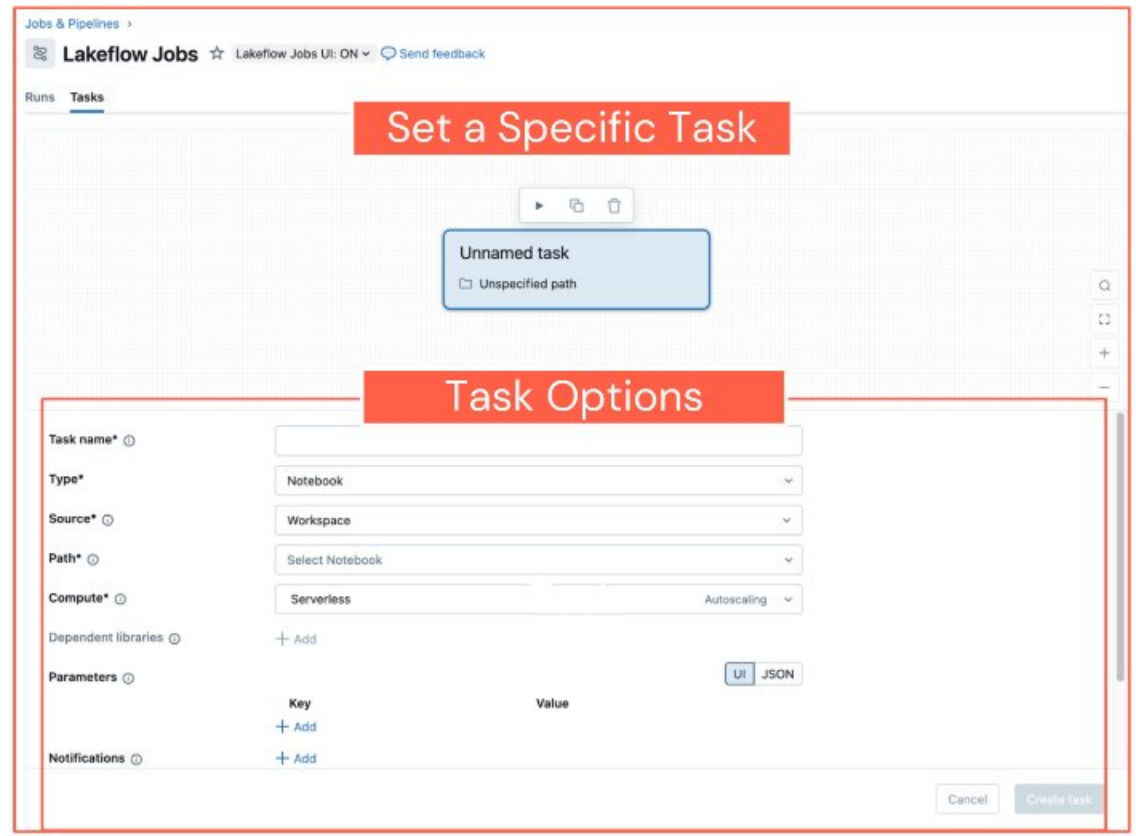


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Building Blocks of Lakeflow Jobs

Tasks Options

- With **Jobs**, we can set a **particular task**
- We have different **options available** based on the task:
  - Defining path
  - Adding libraries
  - Adding parameters
  - Enabling notification
  - All options are specific to the specific task



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Building Blocks of Lakeflow Jobs

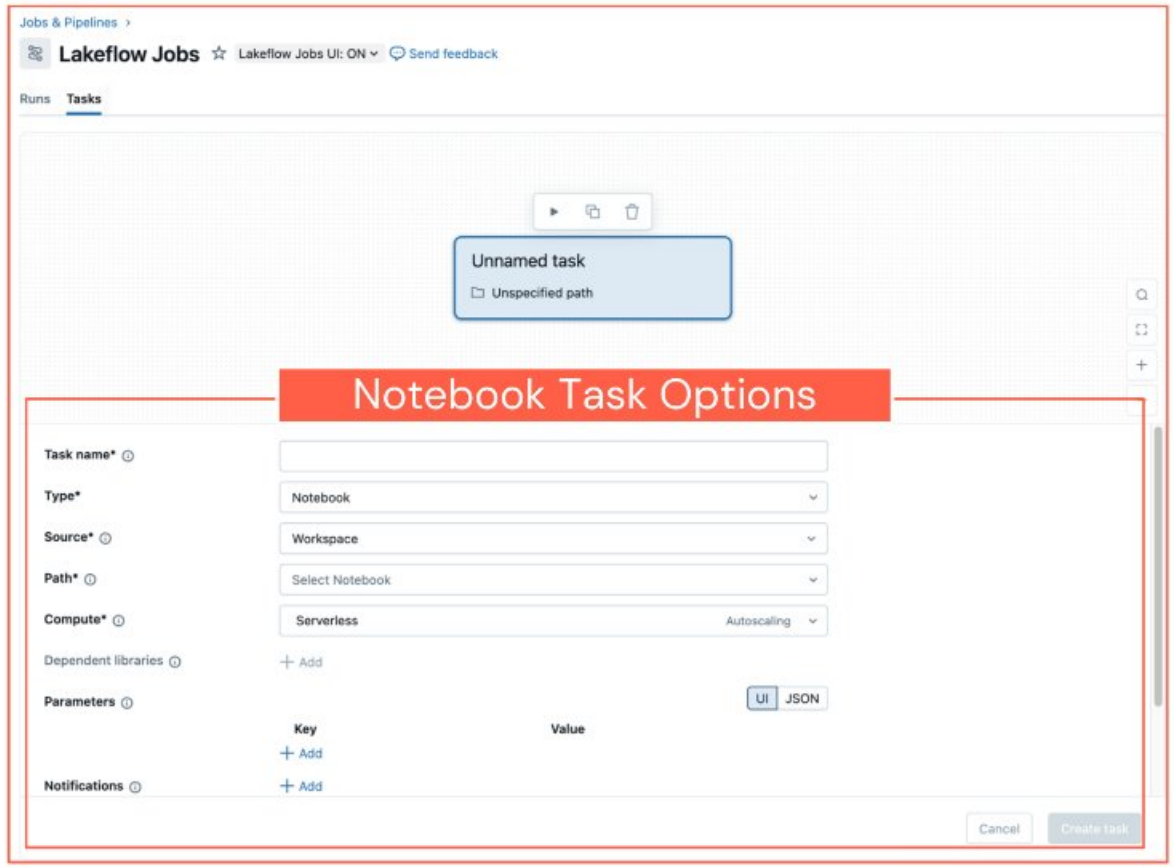
Tasks - Notebook

Based on **task type selected,** you get specific options.

Example

For **Notebook** Tasks:

- Source
- Path
- Compute options
- And More



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Building Blocks of Lakeflow Jobs

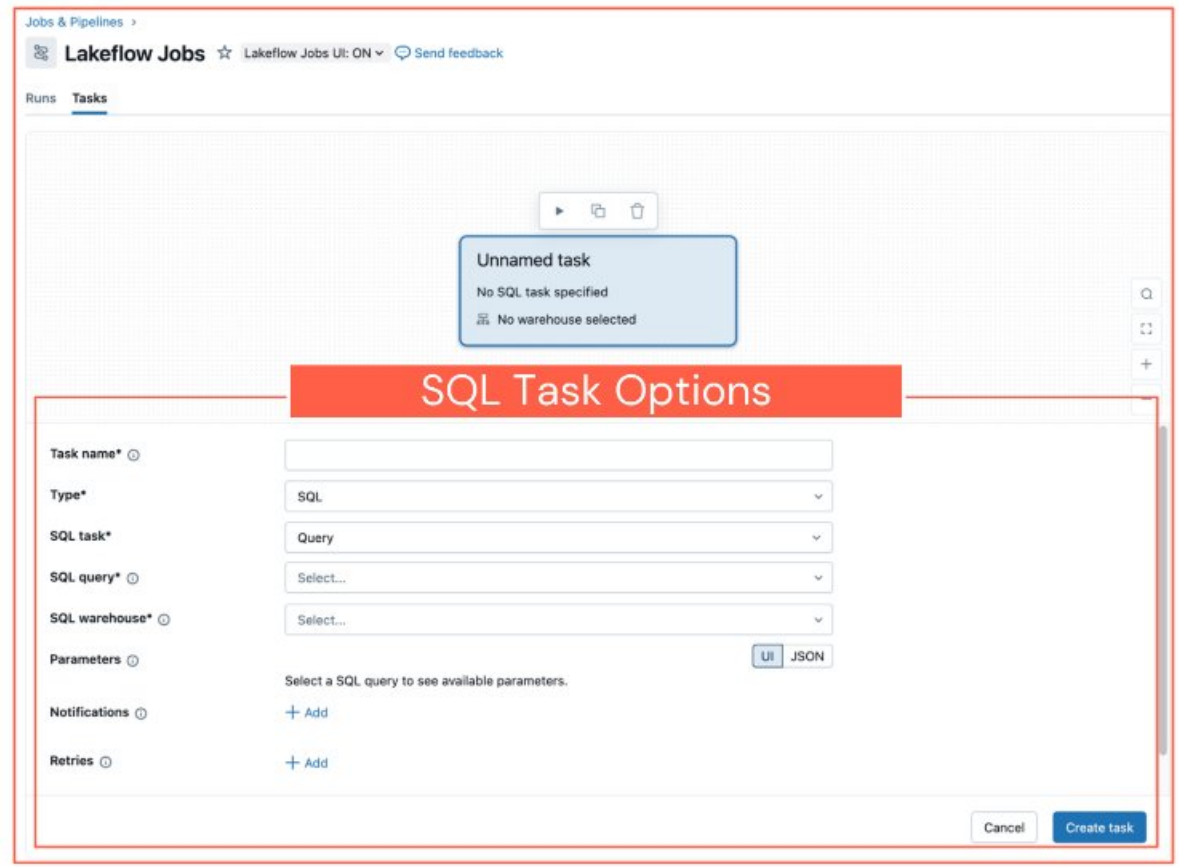
Tasks - SQL

Based on **task type selected,** you get specific options

Example

For **SQL** Tasks:

- Task name
- SQL query
- SQL warehouse



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Building Blocks of Lakeflow Jobs

Languages Supported

- Python
- SQL
- Scala
- R
- Java [via a JAR file]











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Building Blocks of Lakeflow Jobs

Jobs consist of one or more **Tasks**


Databricks Notebooks


DLT


SQL Files/ Queries


Python Scripts


Python Wheels


AI/BI Dashboards


Spark Submit


dbt


Java JAR file


Power BI

Control flows can be established between **Tasks**


Sequential


Parallel


Conditionals (If/else)


Run Job (Modular)


For each

Jobs supports different **Triggers**


Scheduled (Cron)


API Trigger


File Arrival Triggers


Table Triggers


Continuous (Streaming)


Manual Trigger

COMPUTE

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Building Blocks of Lakeflow Jobs

Jobs can be executed on different compute

Development, Ad-Hoc Analysis & Exploration

Interactive Clusters



Interactive or all-purpose clusters can be shared by multiple users.

It is best for performing ad-hoc analysis, data exploration, or development.

Interactive should not be used in Production as they are not cost-efficient.

Production Grade & Operational Use-Cases

Job Clusters



Job clusters are approximately 50% cheaper as they terminate when the job ends, reducing resource usage and costs.

However, Job clusters are subject to cloud providers start-up times.

With Databricks Jobs, you can re-use the same cluster across Tasks for better price performance!

Simple, Fast, Reliable, Cost Efficient!

Serverless



Serverless Workflows are a fully managed service that are operationally simpler and more reliable

They provide you with faster clusters & auto-scaling capabilities providing you with a better user experience for a lower cost.

With out of the box performance optimizations, Serverless provides you with a lower overall TCO.

Compute for SQL Queries and BI, Serverless by default

SQL Warehouse



Purpose-built for SQL queries, dashboards, and BI; Can be attached to notebook as well.

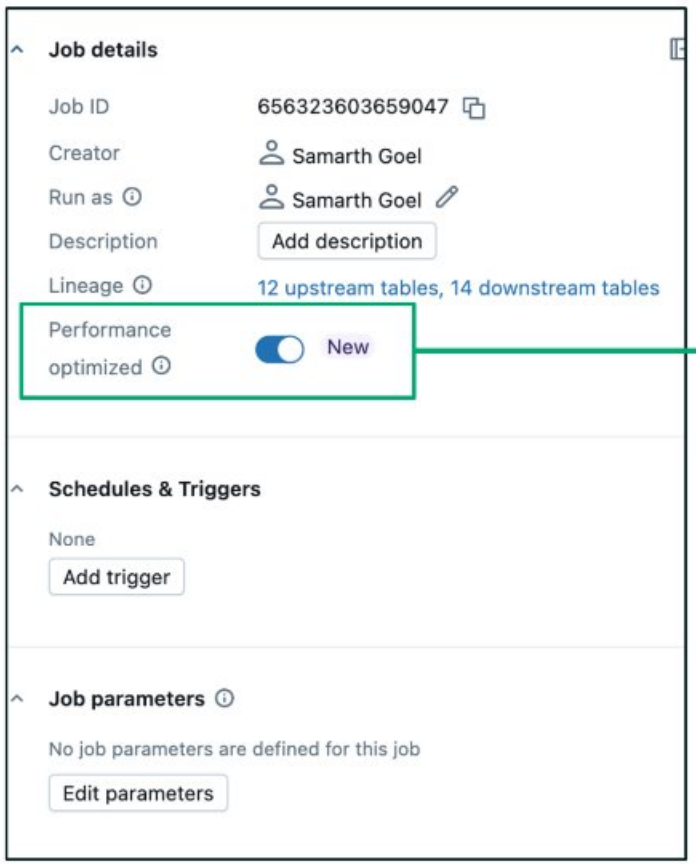
High concurrency + autoscaling via Intelligent Workload Management for consistent low latency.

auto-start/auto-stop and adjustable cluster size and max clusters for peak loads helps control costs.

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Building Blocks of Lakeflow Jobs

Serverless Performance Mode



Use the **Performance optimized** setting in the job details page to choose between **lower cost** and **faster execution** for serverless tasks.

Performance Optimized (Off)

- Standard mode focuses on **cost-efficiency**
- **Longer startup** time (typically 4–6 minutes)
- Best for **non-urgent workloads** with flexible timing

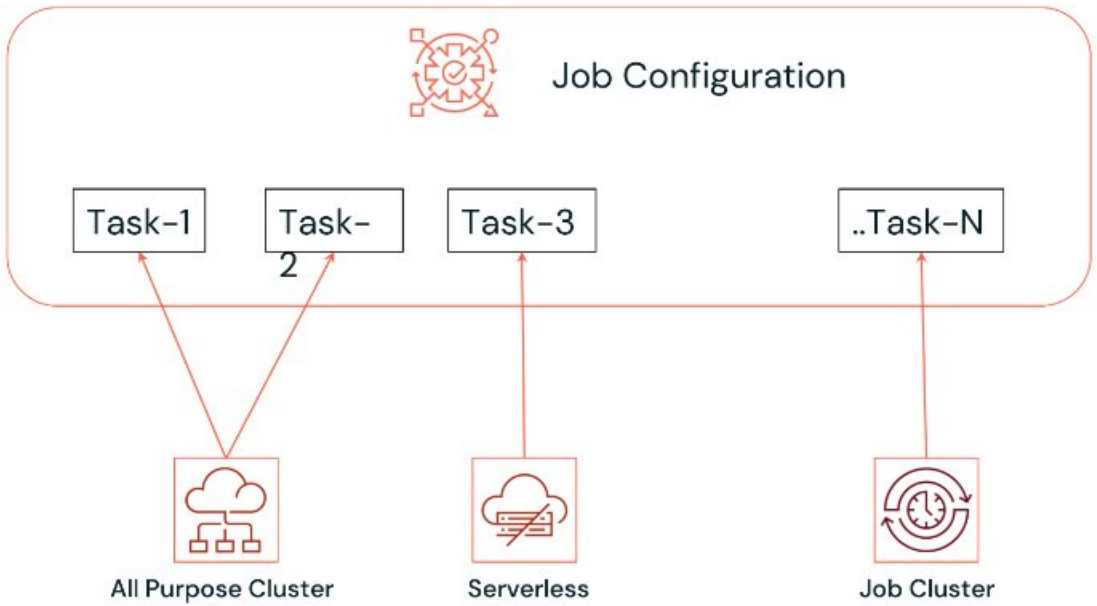
Performance Optimized (On)

- **Enables faster** job startup and execution
- Ideal for **time-sensitive** workloads
- **Applies** only to tasks with **serverless** compute in job

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Building Blocks of Lakeflow Jobs

Selecting Compute



- A Job can have **one or more Tasks** inside it
- Each Task can be assigned its **own compute resource**
- Tasks in the same Job can either:
  - **share** the same compute or
  - use **different** compute as required

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Lakeflow Jobs Core Components

LECTURE

Task Orchestration



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Task Orchestration

What is a DAG?

A DAG is a **conceptual representation** of series of activities, including data processing flows.

- **D**irected - unambiguous direction for each edge
- **A**cyclic - contains no cycles
- **G**raph- collection of vertices connected by edges.

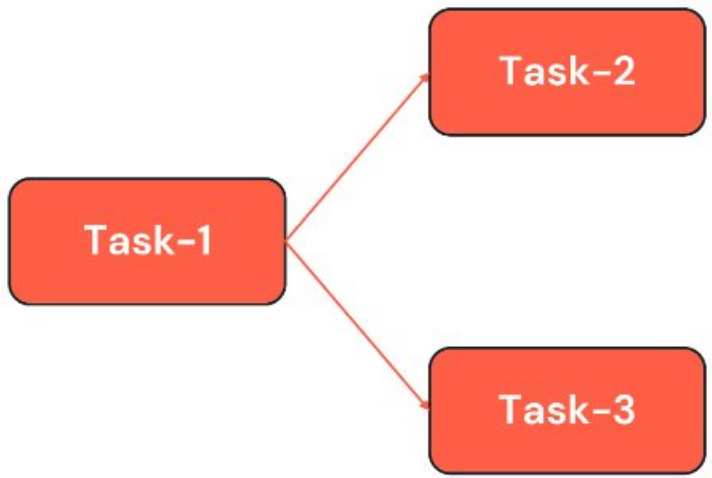


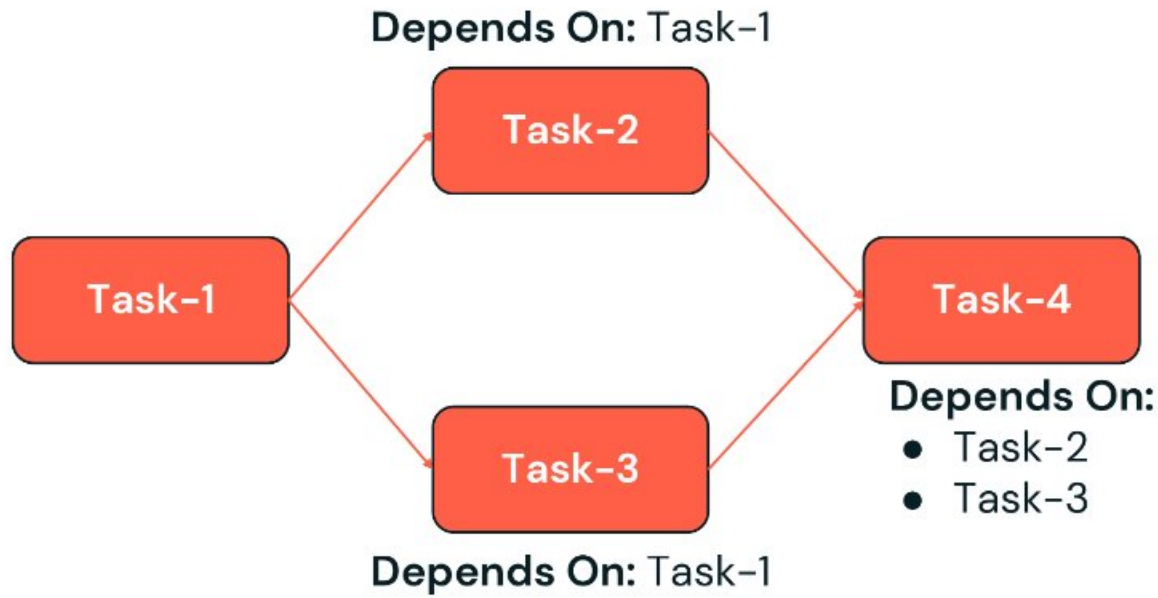
Diagram of a DAG = vertices
edges =

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Task Orchestration

Task Orchestration Overview

- Databricks Jobs supports **task orchestration**
  - The ability to run multiple tasks as a Directed Acyclic Graph (DAG)
  - Orchestrate tasks using the **Databricks UI , API, SDK** or **Databricks Asset Bundles**
- You can define the **order of execution of tasks** in a job by configuring task dependencies, creating a **DAG of task execution**

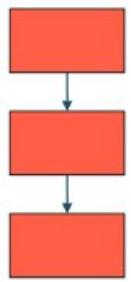


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Task Orchestration

Common Workloads Patterns

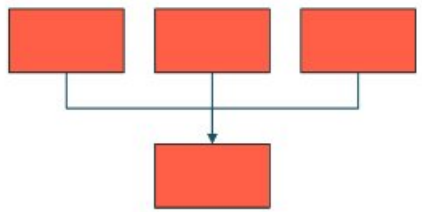
Sequence



Sequence

- Data transformation/ processing/cleaning
- Bronze/silver/gold tables

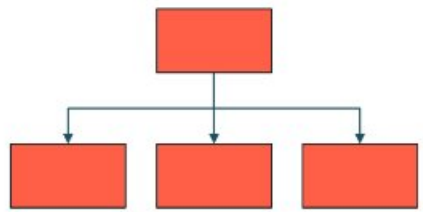
Funnel



Funnel

- Multiple data sources
- Data collection

Fan-out



Fan-out, star pattern

- Single data source
- Data ingestion and distribution

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Introduction to Data Engineering in Databricks

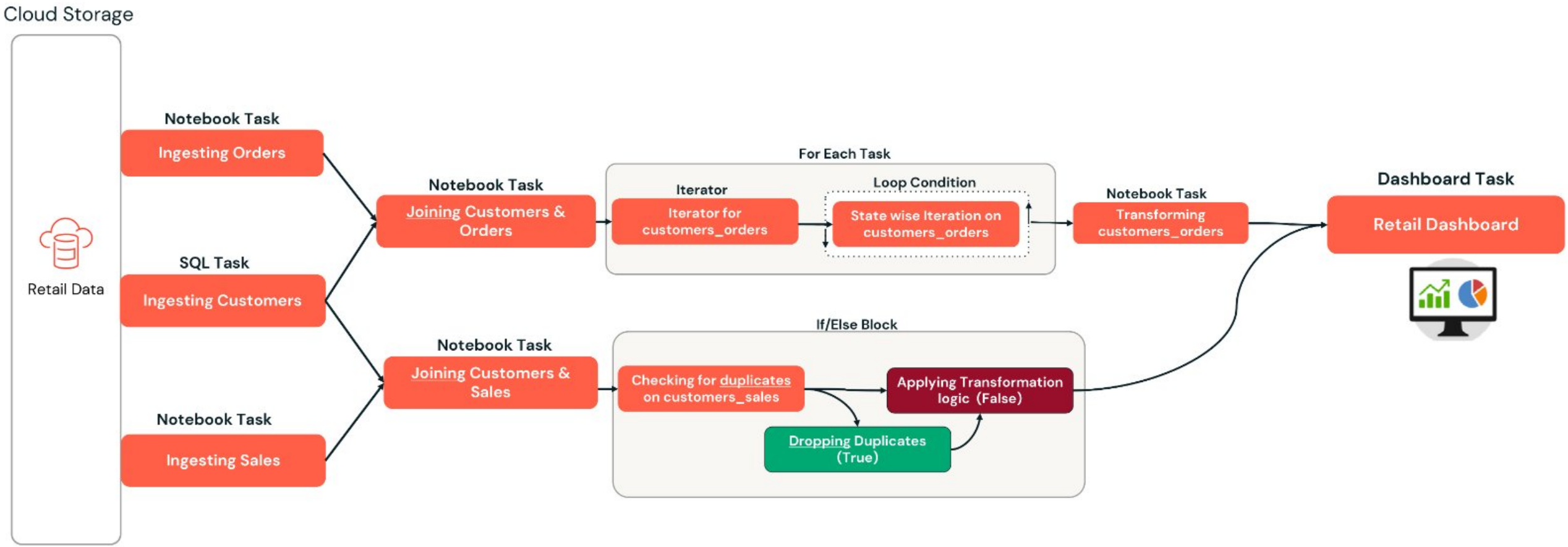
LECTURE

Course Project Overview



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Course Project Overview



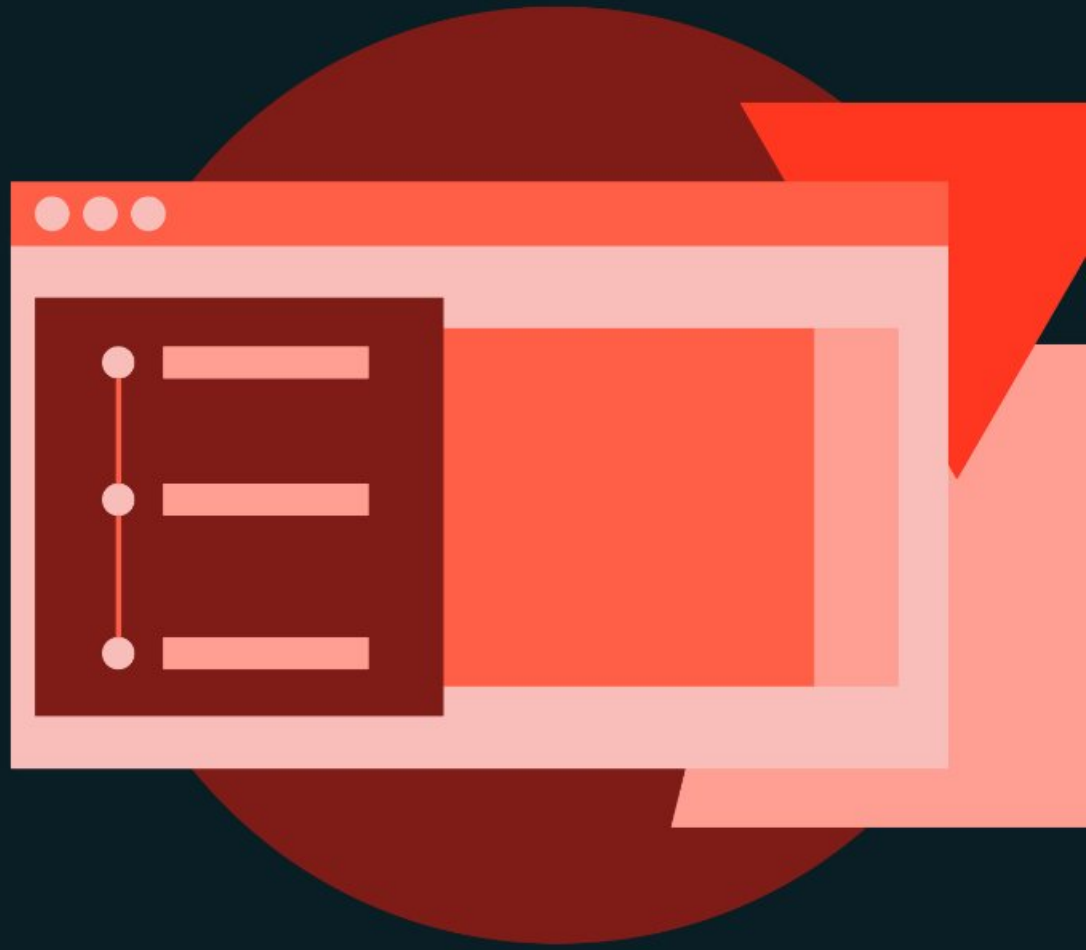
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Lakeflow Jobs Core Components

DEMONSTRATION

Creating a Job Using the Lakeflow Jobs UI



1 Demo – Creating a Job Using the Lakeflow Jobs UI

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Lakeflow Jobs Core Components

LAB EXERCISE

Create your First Job

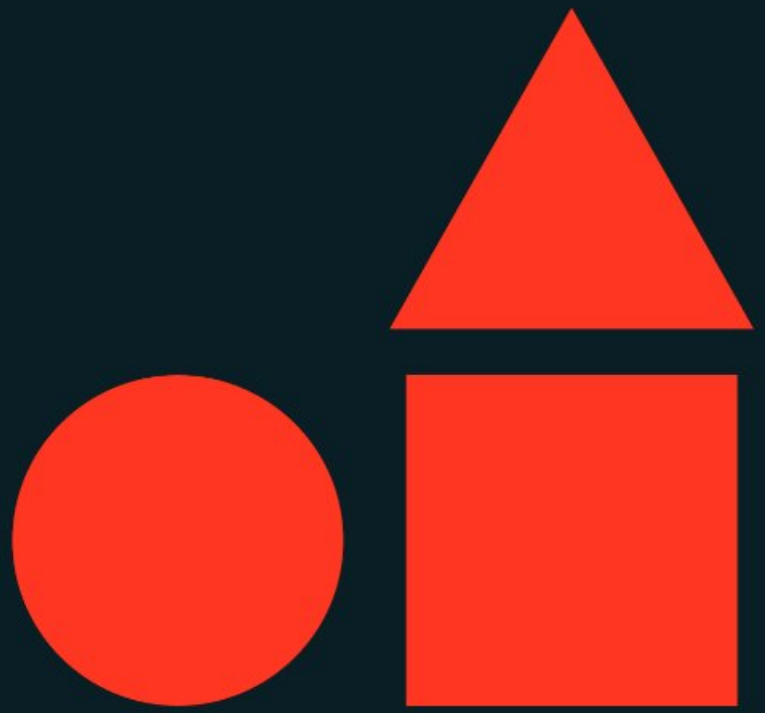


2 Lab - Create your First Job

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Creating and Scheduling Jobs



Deploy Workloads with Lakeflow Jobs

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Section Learning Objectives

- Configure task parameters,task values,dynamic value references, notifications, and retry policies for Lakeflow Jobs.
- Set up and manage different job scheduling options, including scheduled, file arrival, Table update and continuous triggers.
- Automate job execution using triggers to support time-based and event-driven workflows.
- Explore and apply scheduling options in the Databricks Lakeflow Jobs UI through hands-on demonstration.



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Agenda

Creating and Scheduling Jobs
• Common Task Configuration Options
• Jobs Schedulers and Triggers
• Demo – Exploring Scheduling Options



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Creating and Scheduling Jobs

LECTURE

Common Task Configuration Options



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Common Task Configuration Options

Overview



Parameters & Dynamic Value References



Notification Alerts



Retries

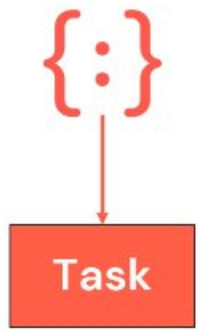
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Common Task Configuration Options

Parameter Overview

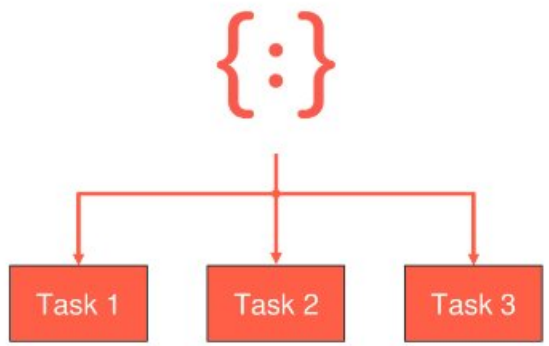
Task Parameters

A key-value pair defined at the **task level**



Job Parameters

A key-value pair defined at the **job level** and pushed down to all tasks



Job Parameters **override** Task Parameters when same **key** exists

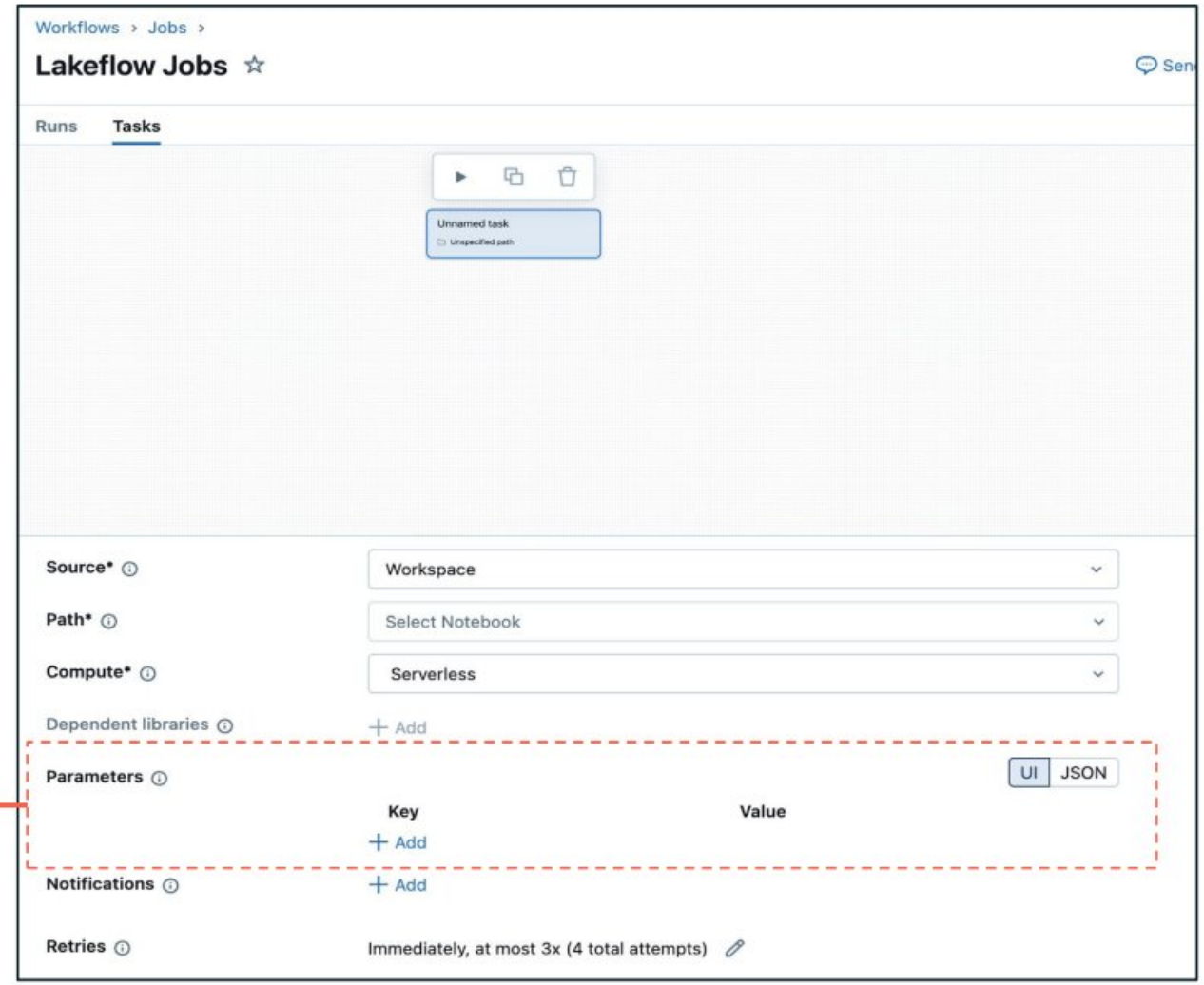
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Common Task Configuration Options

Task Parameter Overview

- **Task Parameters** are **key-value** pairs that lets you pass values into tasks
- Support **advanced orchestration** such as
  - Conditional execution
  - Looping
  - Passing context between tasks





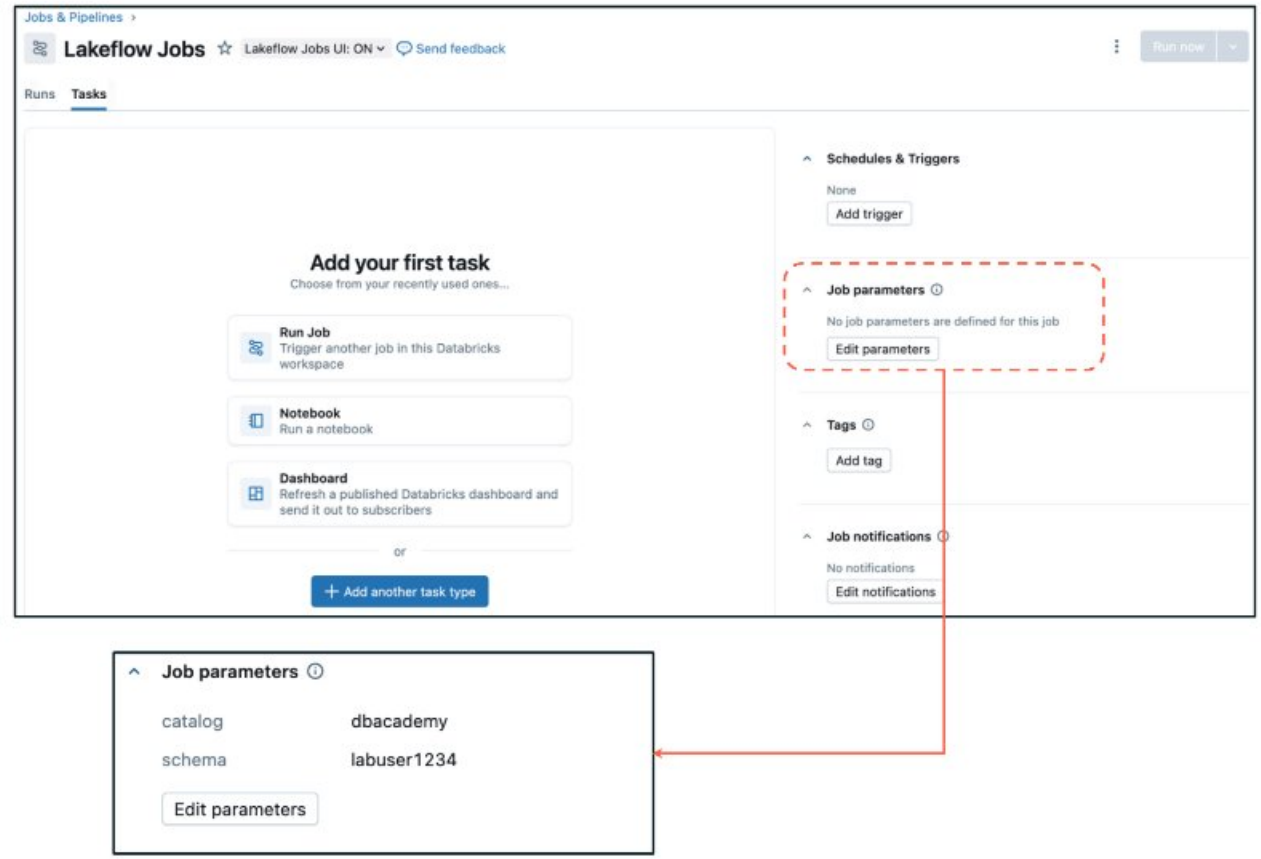
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Common Task Configuration Options

Job Parameter Overview

Job Parameters are **key-value pairs** defined at the **job level** that provide default values for the entire workflow.

- Applied to **all tasks** in the job automatically
- **Override task parameters** with the same key name
- Can be **overridden at runtime** when triggering job runs



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Common Task Configuration Options

Setting and Accessing Parameters (UI)

Setting Parameter Values (UI)



Accessing Parameter Values in a Task

```
catalog = dbutils.widgets.get("catalog_name")

schema = dbutils.widgets.get("schema_name")
```

Setting Job Parameters:

- In your **job**, go to the **Parameters section**, and add key–value pairs at the job level.

Setting Task Parameters:

- Go to the **specific task in the job**,
- Go to **parameters** within the **task configuration** (alongside task name, type, and path)
- Add the key–value pairs

- Access both **task and job parameters** in notebook tasks with **`dbutils.widgets.get`**
- Retrieval methods vary by **task type** (e.g., notebook vs. SQL vs. Python wheel)

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Common Task Configuration Options

Dynamically Setting and Accessing Task Parameters

Dynamically Set Task Values with Code

```
dbutils.jobs.taskValues.set(
                key = "catalog_name",
                value = "dbacademy")
```

Task Values are **dynamic key-value pairs** that tasks create and share during **workflow execution**.

Accessing a Parameter from another Task

```
dbutils.jobs.taskValues.get(
            taskKey="task-name",
                    key="catalog_name")
```

You Reference specific upstream task parameter values into **downstream tasks**



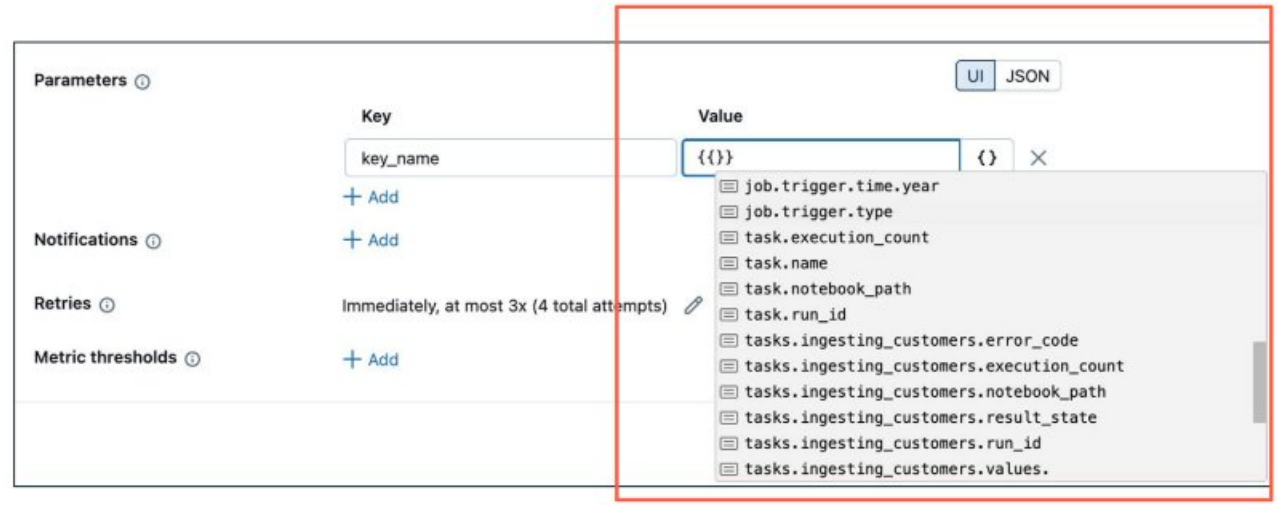
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Common Task Configuration Options

Dynamic Value References

- **Dynamic Value References** allow referencing values at **runtime** from the job and task context (e.g., task outputs, parameters, retask values, registered task values, run metadata).
- These references use `{{ }}` notation and allow workflows to adapt to different execution.
- Common examples:
  - `{{job.start_time.day}}` - For accessing the day
  - `{{task.name}}` - For Accessing Task Name
  - `{{tasks.<task-name>.values.my_value}}`

    For accessing output from an upstream task



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Common Task Configuration Options

Notification Alerts



Notification Configurations

Job Level Notifications

- A notification alert is sent after **successful completion of the job**
- This setting can be adjusted in the Job Details section (right-side pane).

Task Level Notifications

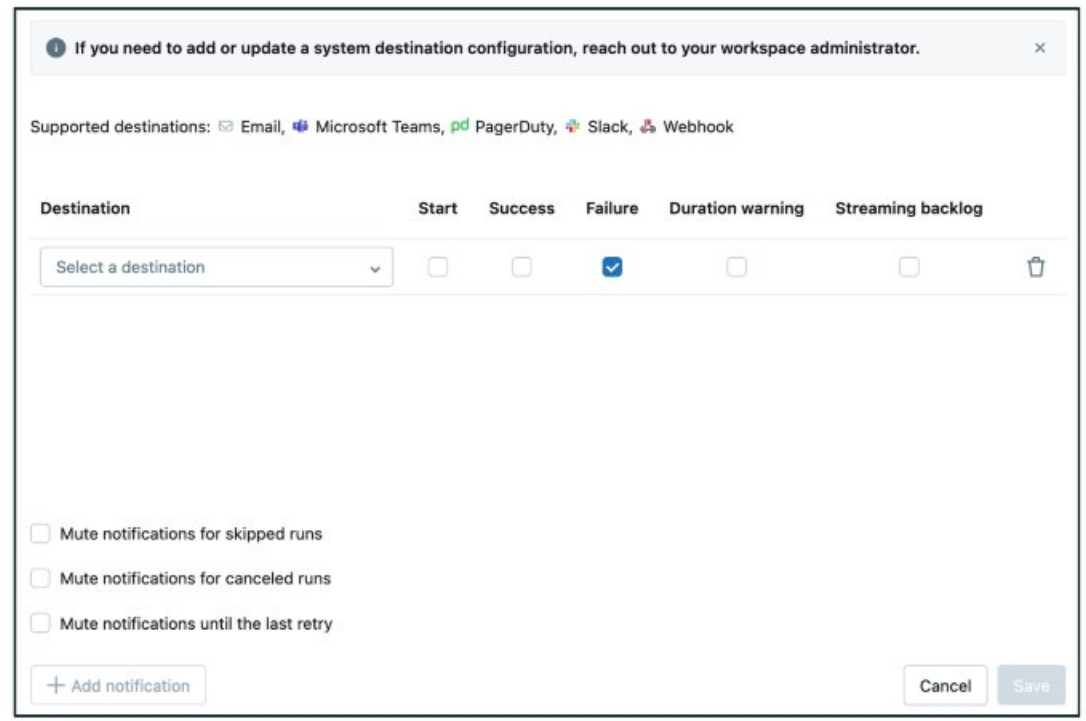
- A notification alert will be sent when the **task is successfully completed**
- This notification setting can be customized at the task level.

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Common Task Configuration Options

Notification Alerts

- **Multiple** supporting destinations as Emails, Teams, pagerDuty,Slack and Webhook
- **Each task in a Job** could be configured **differently** to send notification
- Jobs triggers **notifications** when the task begins, completes or fails
- These notification could also be triggered in the case of **late Jobs:**
  - For duration threshold warning/timeout alerts
  - Streaming backlog(Issue with getting streaming data)
- Webhook allows custom integration with third-party API services

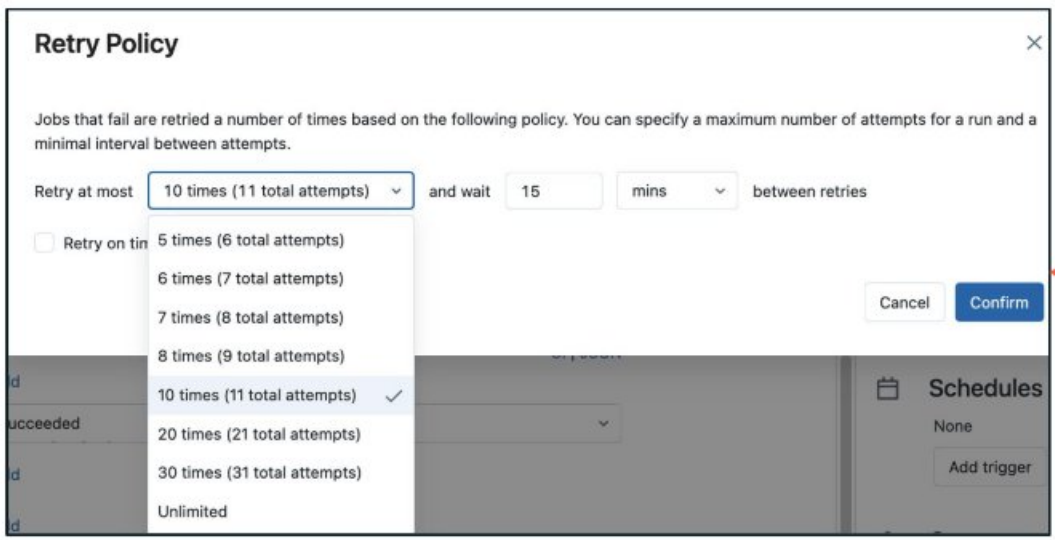


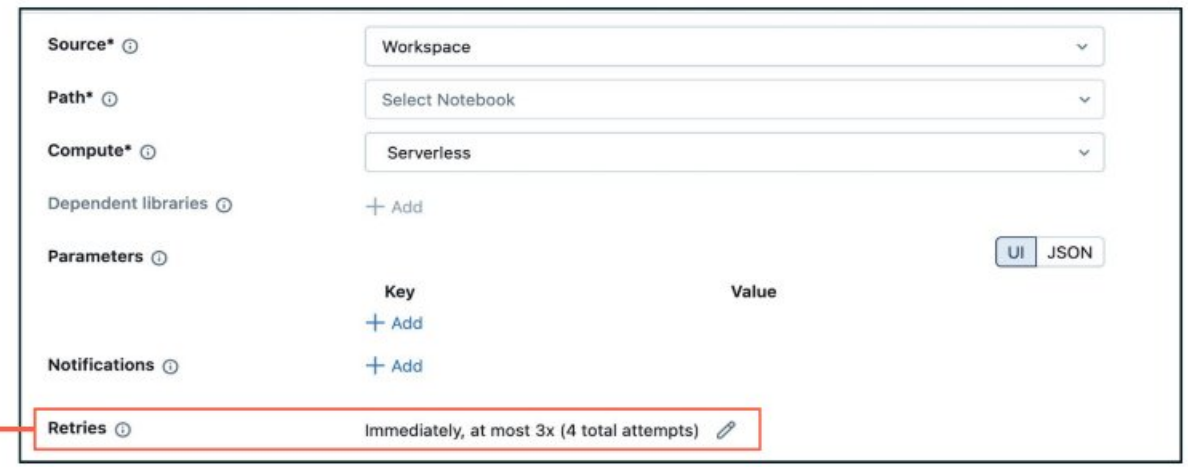
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Common Task Configuration Options

Retry Policy

A policy that determines when and how many times failed runs are **retried**





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Creating and Scheduling Jobs

LECTURE

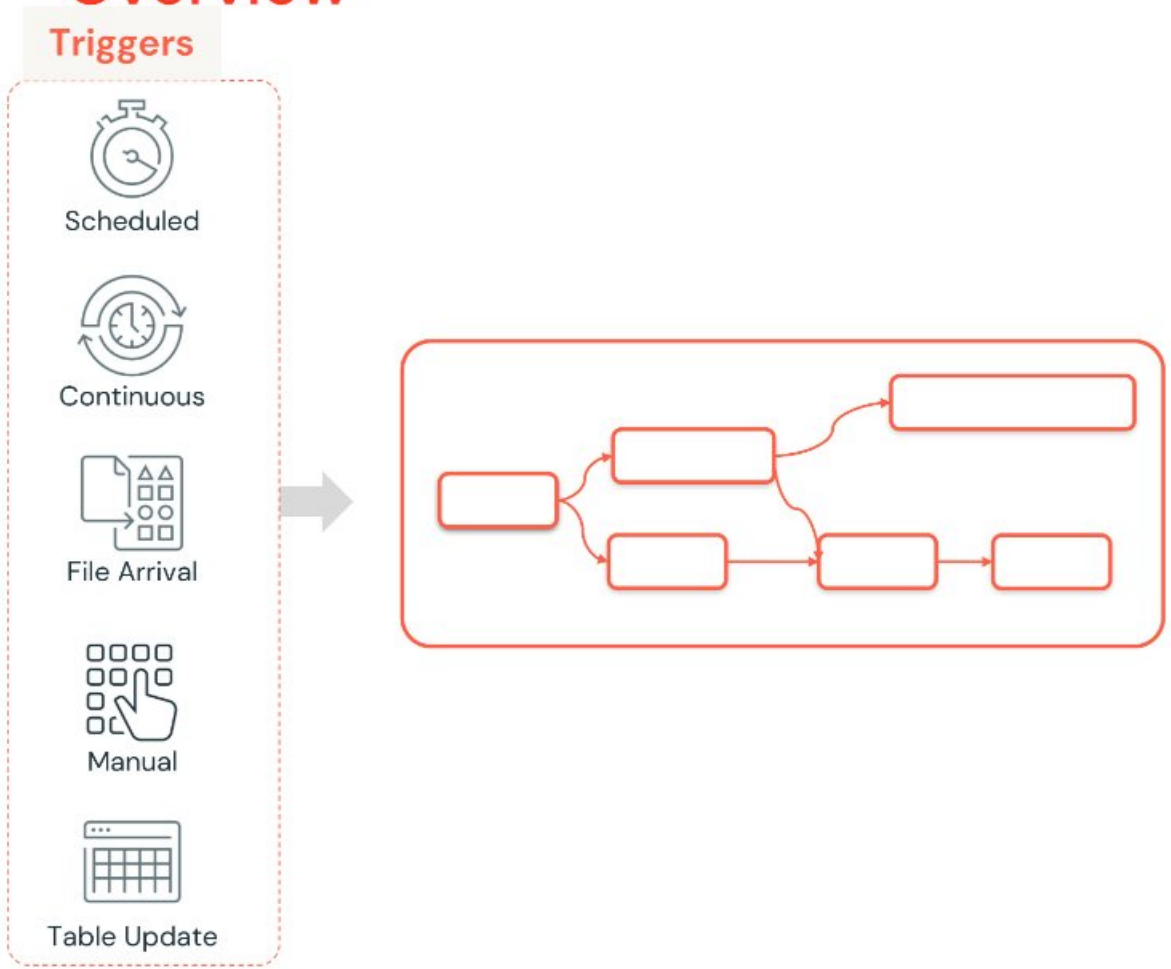
Job Schedules and Triggers



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Job Schedules and Triggers

Overview



- A **trigger** is a rule that automatically starts a job run based on a **specific condition** or **schedule**
- **Common** trigger types include:
  1. Time-based schedules
  2. Continuous (always-on) execution
  3. File arrival events
  4. Manual trigger
  5. Table Update
- Triggers **enable automation**, so jobs can run without manual intervention

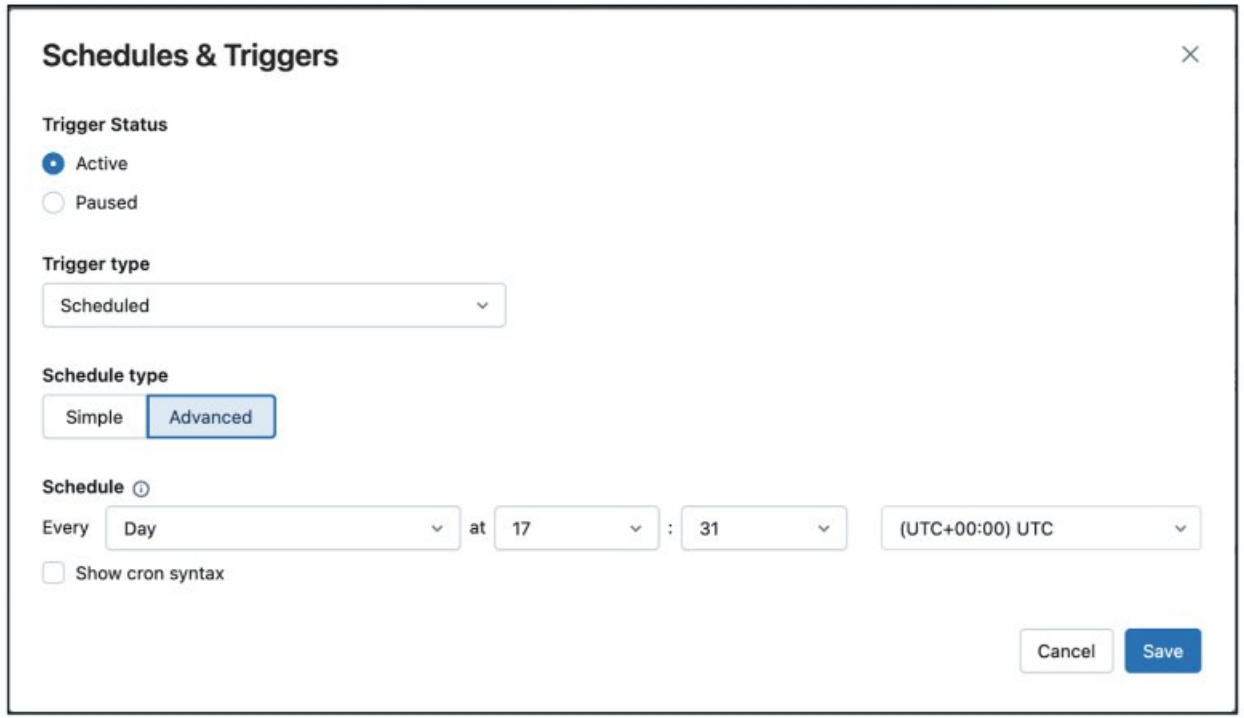
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Job Schedules and Triggers

Types of Triggers

1. Scheduled Trigger

- **Automatically run jobs** at set times or intervals, such as hourly, daily using the UI
- You can schedule jobs using **cron expressions** for automated execution
- They help **automate** recurring jobs, ensuring jobs run consistently without manual intervention



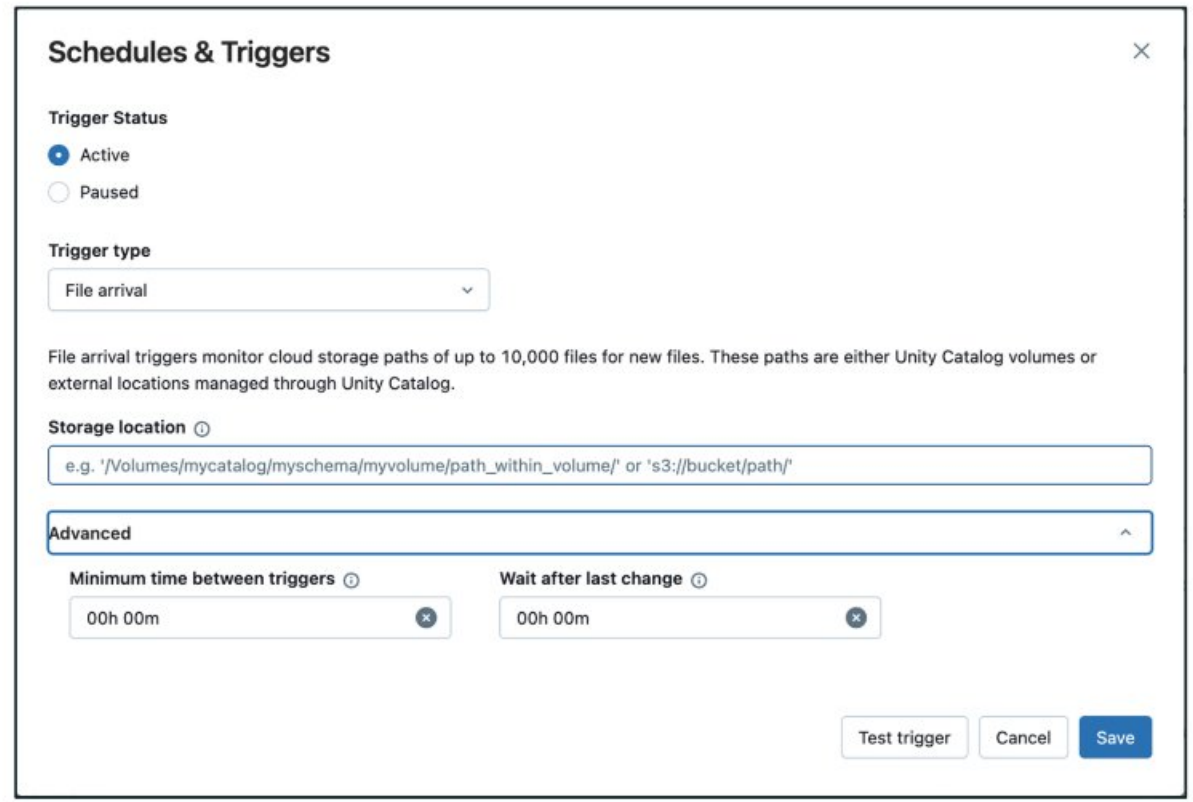
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Job Schedules and Triggers

Types of Triggers

2. File Arrival Trigger

- **Automatically** trigger jobs **when new files are detected** in a specified storage location, with support for:
  - AWS S3, Azure Storage, GCP GS and Databricks Volumes
- Enable **event-driven processing** to start data workflows upon **file arrival**
- Ideal for **automating jobs** with **unpredictable or irregular** data ingestion patterns



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Job Schedules and Triggers

Types of Triggers

3. Continuous Trigger

- **Continuously** runs jobs by starting a new run as soon as the previous one finishes or fails
- **Built-in retry** logic is automatically managed by Databricks
- Ideal for **streaming** workloads



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Job Schedules and Triggers

Types of Triggers

4. Manual Triggers

- **Manual (None) trigger** lets jobs run on demand without any schedule or event.
- Can be **started from UI** (Run now or Run now with different settings), API, CLI, SDK, or via DABS
- Best for **ad hoc runs, debugging, and one-off backfills**
- Can be combined with other trigger types later



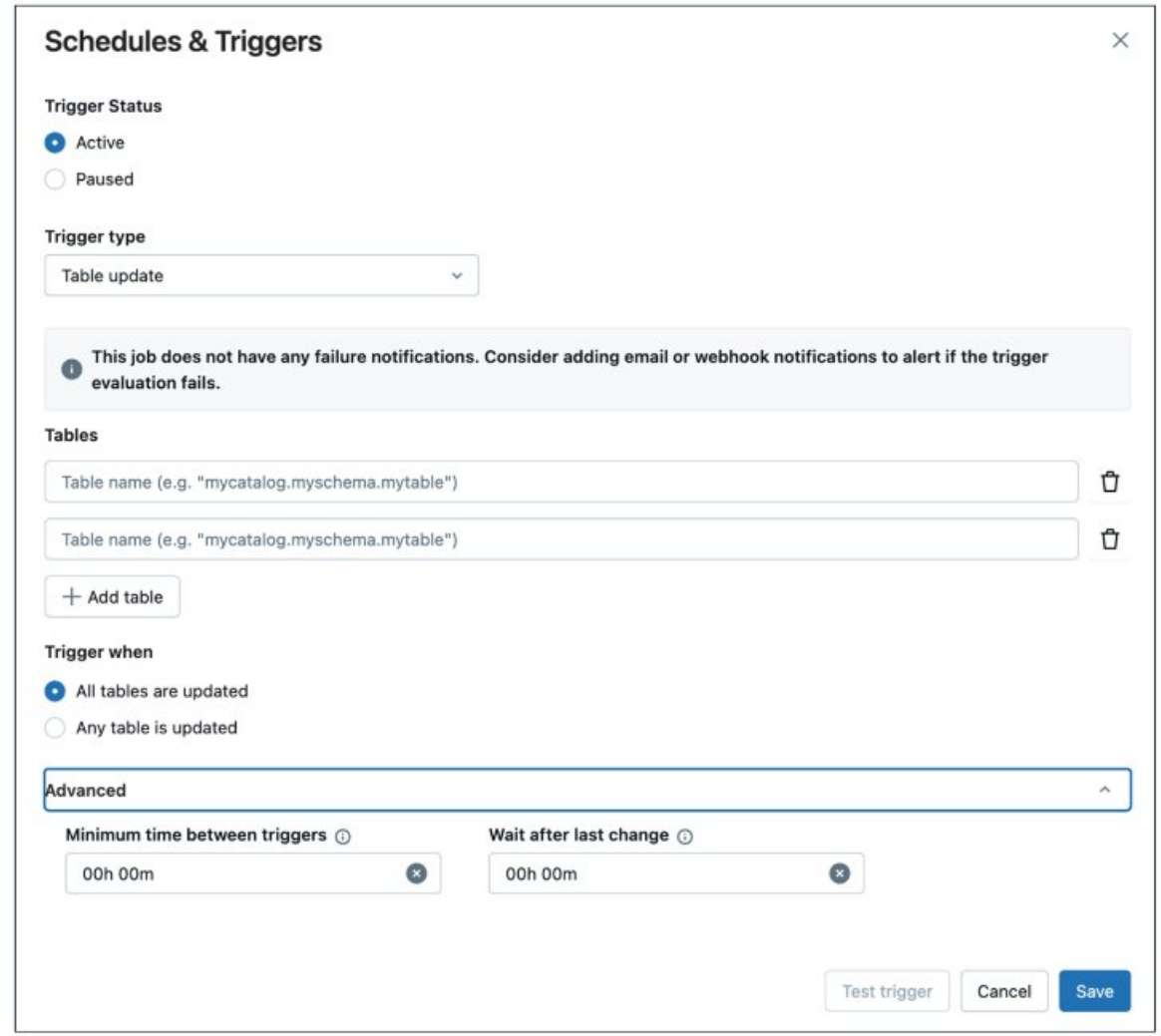
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Job Schedules and Triggers

Types of Triggers

5. Table Update Trigger

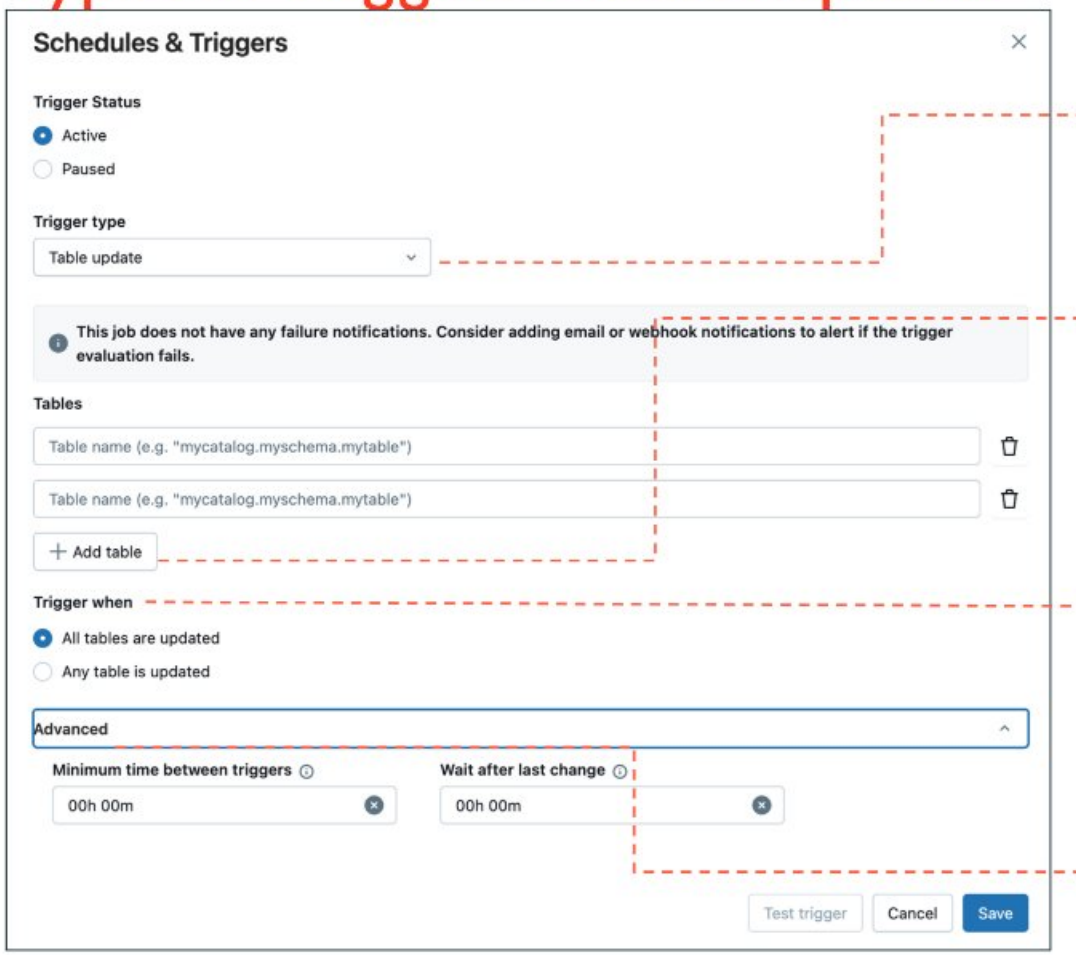
- Table Update Trigger automatically starts a job when **specified tables** are updated .
- **Monitors** one or more tables for changes (insert, update, delete, or merge) .
- Helps **replace** time-based scheduling (like cron jobs) with real-time orchestration, ensuring jobs run as soon as **new data arrives .**



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Job Schedules and Triggers

Types of Triggers - Table Update



Step-1

Selecting trigger type

Step-2

Adding source tables

- You can select up to 10 tables per trigger.
- Works with Unity Catalog-managed Delta, Iceberg, external Delta tables, materialized views, and streaming tables.

Step-3

Defining triggering condition

- Any table updated – triggers when the first monitored table changes.
- All tables updated – triggers only after all selected tables update .

Step-4

Advanced functionality(Optional)

- Min. time between triggers - Prevents over-triggering by enforcing a mandatory gap between consecutive job runs
- Wait after last change - Delays job execution until all data updates have landed in source table

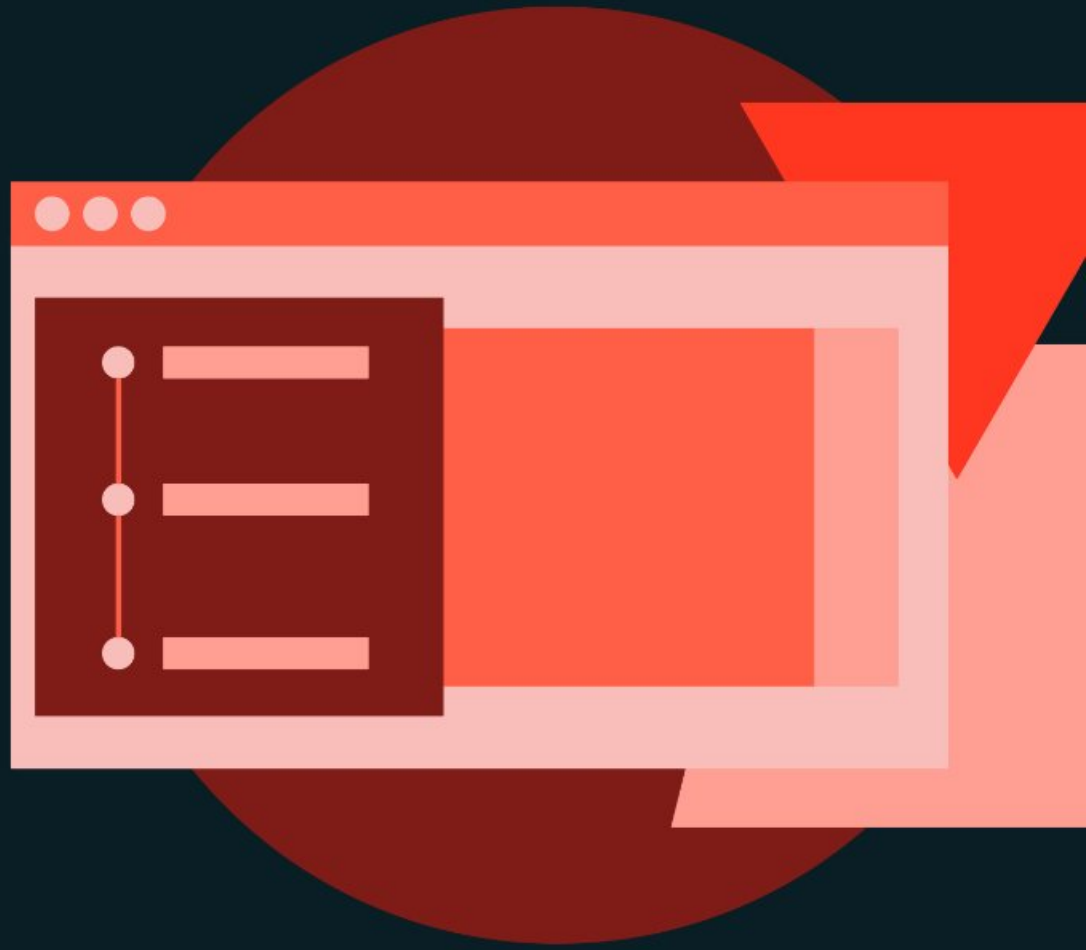
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Creating and Scheduling Jobs

DEMONSTRATION

Automating Workloads with Scheduling and Triggers

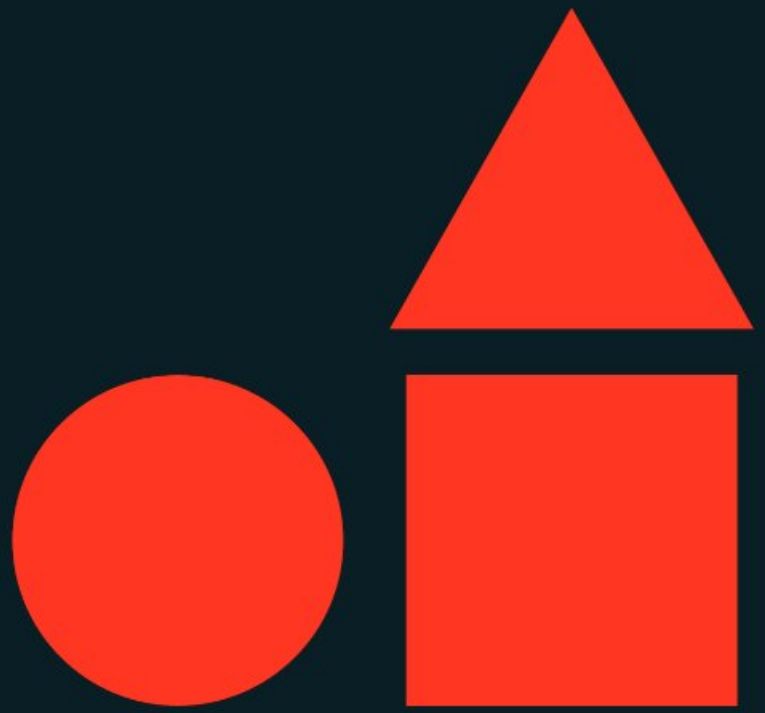


3 Demo - Automating Workloads with Scheduling and Triggers

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Advanced Lakeflow Jobs Features



Deploy Workloads with Lakeflow Jobs

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Section Learning Objectives

- Implement conditional task execution using run-if dependencies and If/Else logic.
- Utilize the repair run feature to efficiently recover from failed job runs.
- Design production-ready architectures with modular orchestration patterns, Git integration, and appropriate compute selection (serverless vs interactive clusters) for scalable workload management.
- Optimize performance monitoring and cost management using system tables (system.lakeflow) and Spark UI insights to identify bottlenecks, track SLAs, and implement resource efficiency measures.
- Apply enterprise best practices including service principal usage, parameterized tasks, alert configuration, and maintainable job design patterns to ensure reliable production deployments.



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Agenda

Advanced Lakeflow Jobs Features
• Conditional and Iterative Tasks
• Handling Task Failures
• Lakeflow Jobs in Production
• Best Practices



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Advanced Lakeflow Jobs Features

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Conditional and Iterative Tasks



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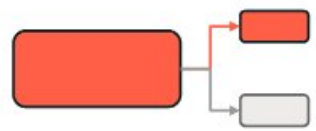
Conditional and Iterative Tasks

Overview

Run if Conditional Task Dependencies



If Else Task



If/else

For Each Task



For each

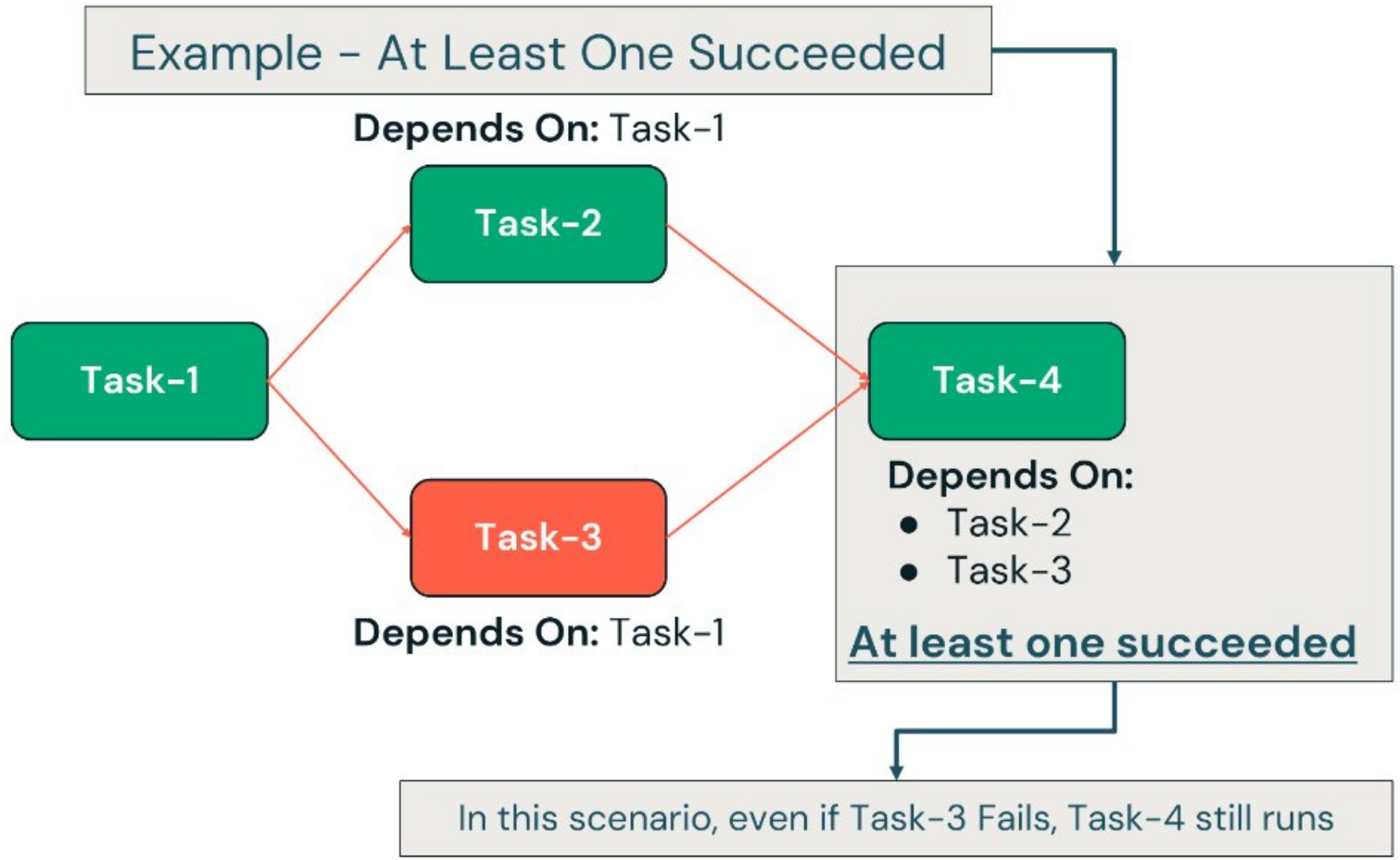
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Conditional and Iterative Tasks

Run if Conditional Task Dependencies

Run if Conditional Task Dependencies

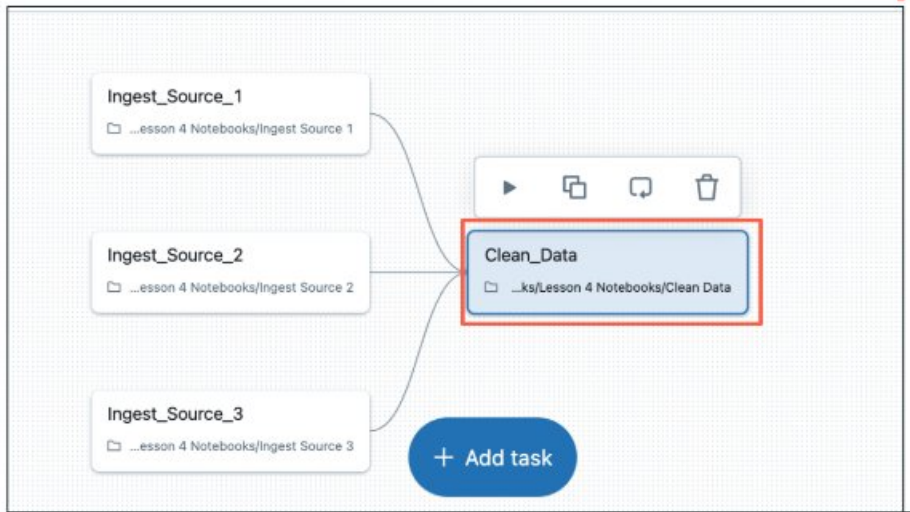
- Control task execution based on outcomes of **upstream tasks**
- Supports a variety of **dependency conditions** like:
  - All succeeded
  - At least one succeeded
  - None failed
  - And more



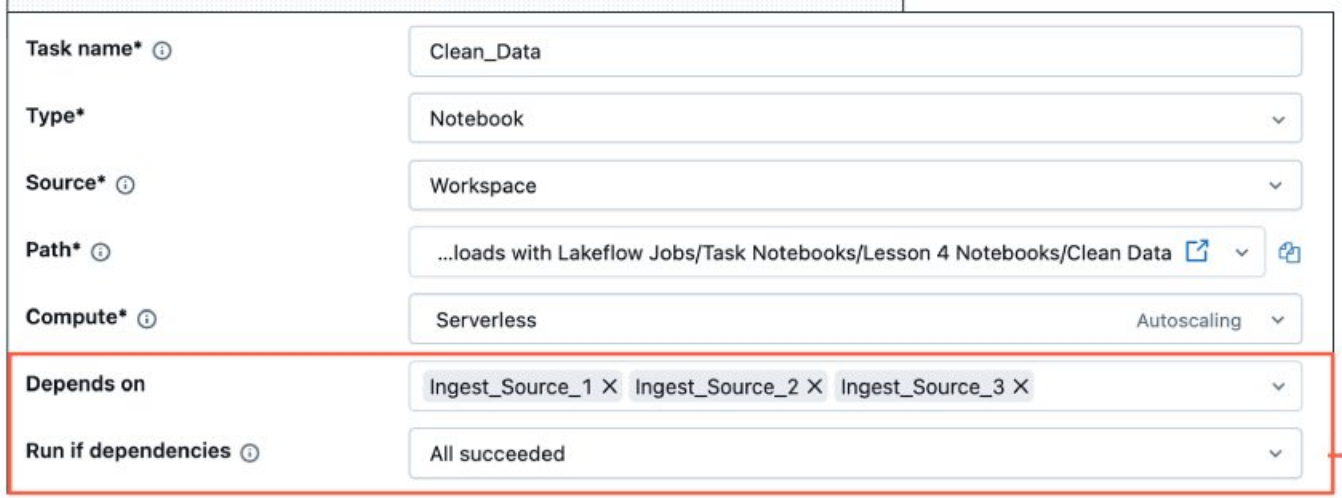
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Conditional and Iterative Tasks

Run if Conditional Task Dependencies



- Task dependencies provide **control flow logic based** on upstream task status
- Visually represented in the job DAG





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Conditional and Iterative Tasks

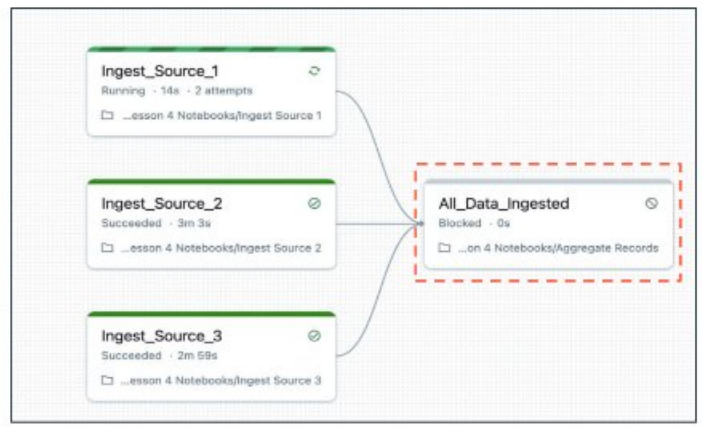
Run if Conditional Task Dependencies

1. **Job Initialization**

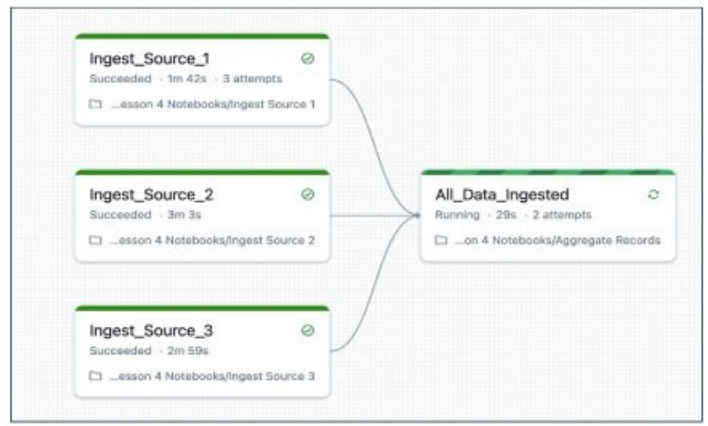
The first three tasks are executed



2. Waiting for all task to complete (*All succeeded*)



3. Dependent Task Started



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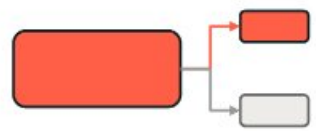
Conditional and Iterative Tasks

If Else Task

Run if Conditional Task Dependencies



If Else Task



If/else

For Each Task

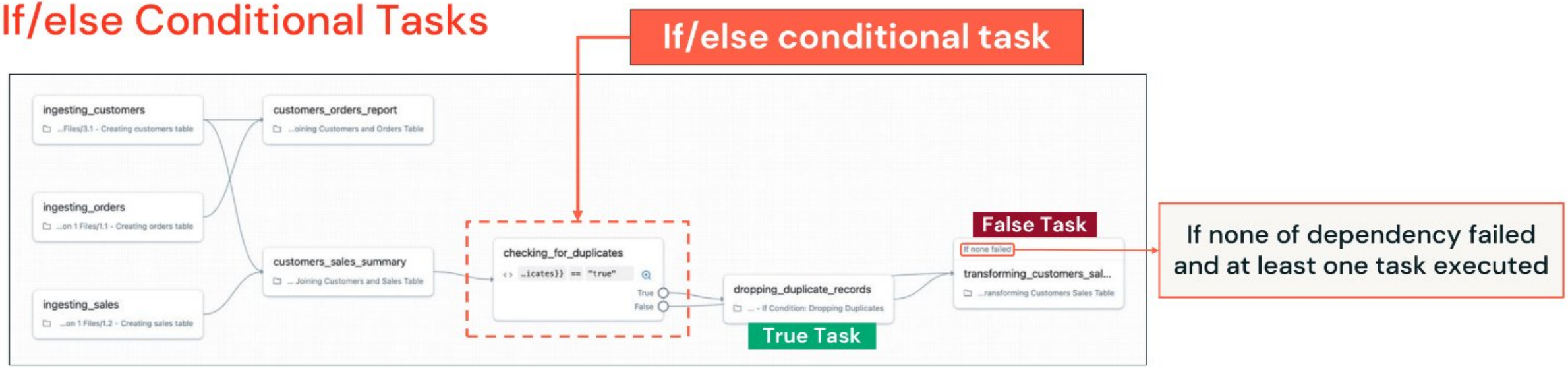


For each

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Conditional and Iterative Tasks

If/else Conditional Tasks



If/else Conditional Tasks

- Adds boolean conditional logic to your workflow based on task results
- Allows branching based on specific conditions (e.g., data quality checks, record counts)
- Uses boolean operators (==, !=, >, >=, <, <=) to evaluate expressions

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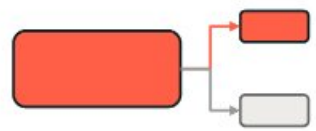
Conditional and Iterative Tasks

For Each Task

Run if Conditional Task Dependencies



If Else Task



If/else

For Each Task

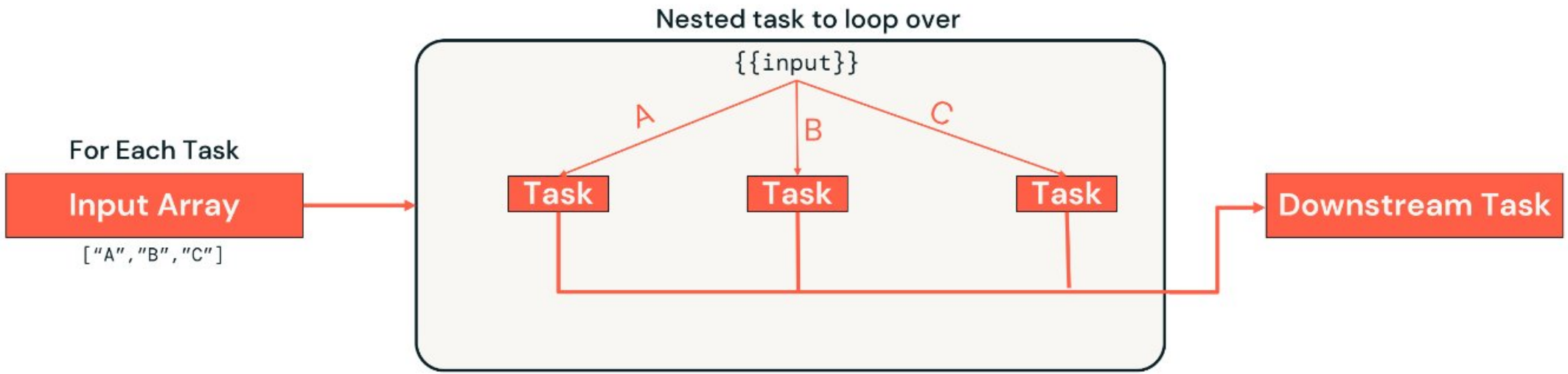


For each

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Conditional and Iterative Tasks

For Each Task



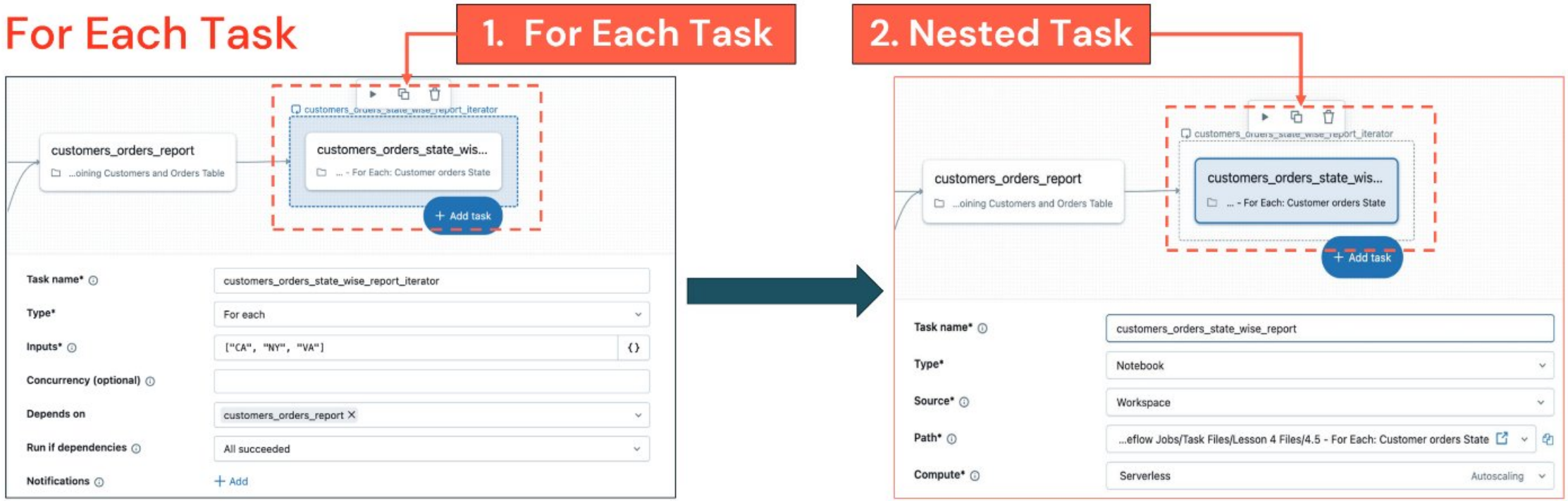
For Each Task

- For Each loops over an input array and runs the same nested task per item, passing it as `{{input}}`
- Iterations can run in parallel with configurable concurrency to speed up execution.
- Downstream dependencies attach to the For Each container, not to the nested task.

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Conditional and Iterative Tasks

For Each Task



For Each task requires defining two components:

1. **The For Each task -** The top-level container task that manages the loop
2. **A nested task -** The actual task that executes for each iteration

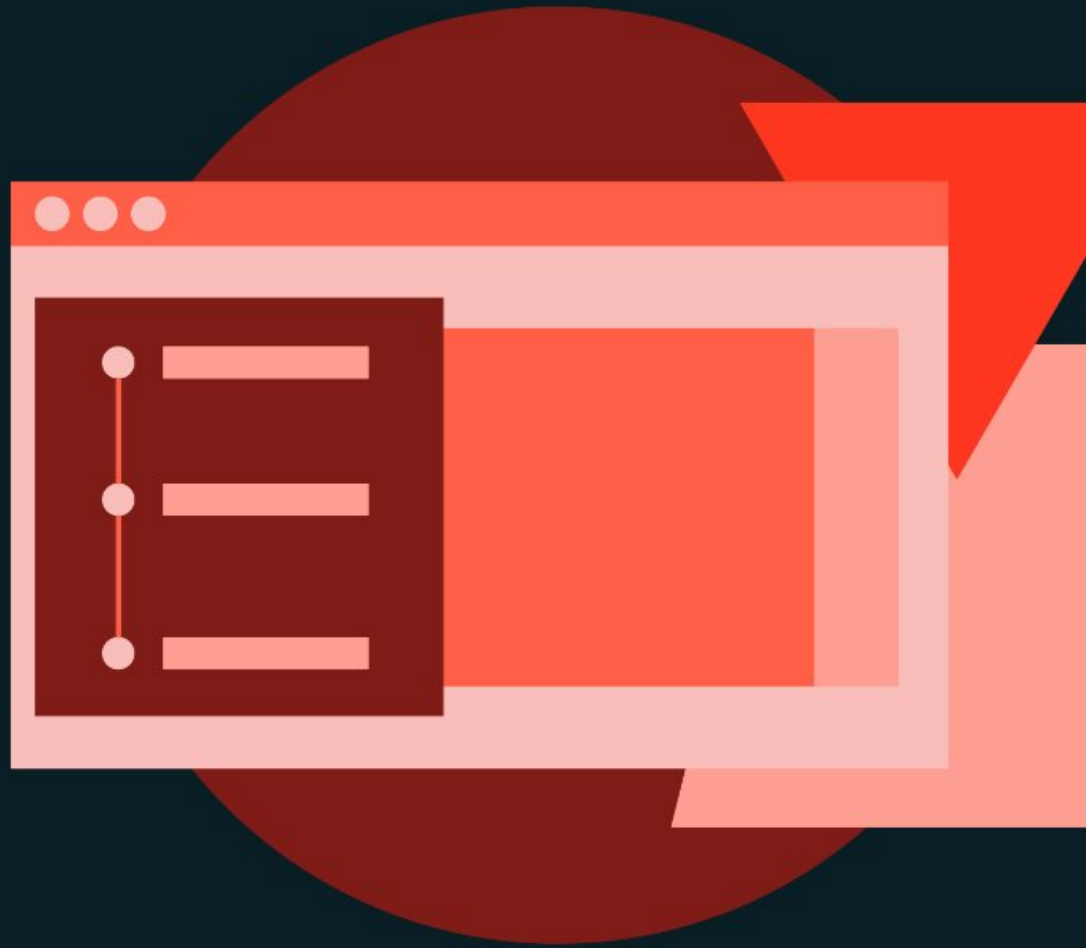
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Advanced Lakeflow Jobs Features

DEMONSTRATION

Building Dynamic Workloads with Advanced Tasks



4 Demo - Building Dynamic Workloads with Advanced Tasks

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Advanced Lakeflow Jobs Features

LAB EXERCISE

Adding If-Else Task and Automating your Job



5 Lab - Adding If-Else Task and Automating your Job

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Advanced Lakeflow Jobs Features

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Handling Task Failures



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Handling Task Failures

Repair and Rerun



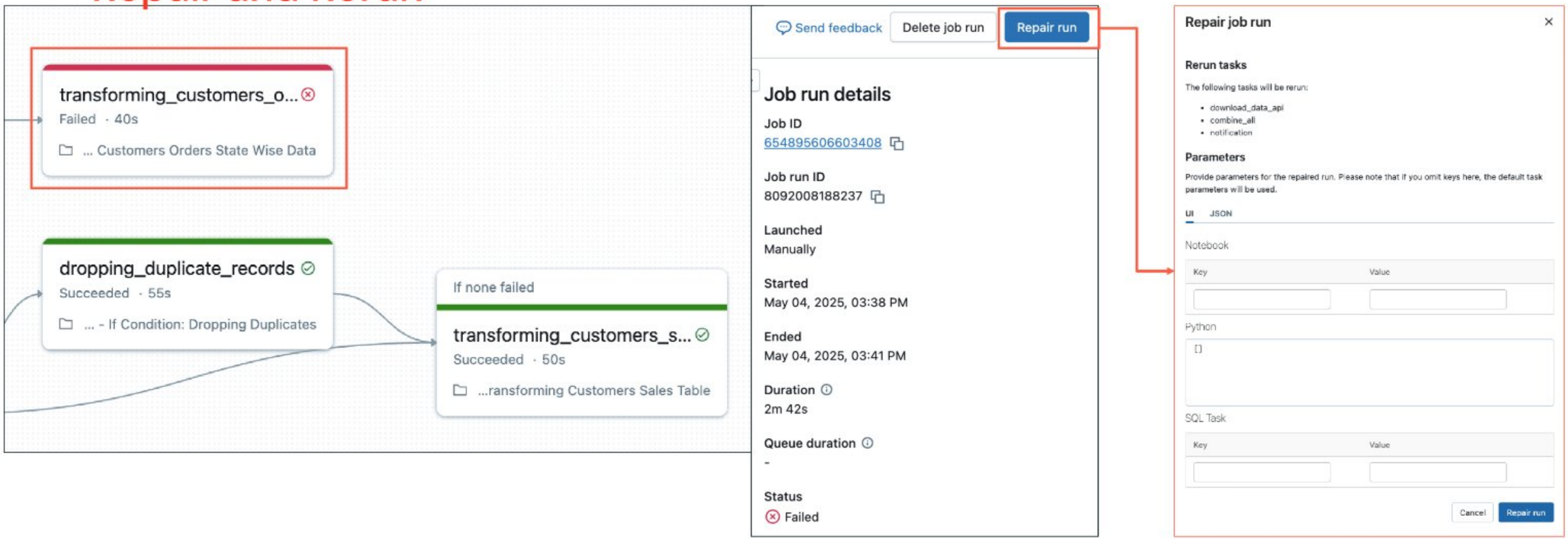
Re-run the **failed** task

- **Repair** feature allows to **re-run** and override task parameters
- **Reduces the time** and resources required to recover from unsuccessful job runs
- In case of **Task Failure,** You can:
  - Modify the **task** and run again
  - Modify the **parameters** and run again

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Handling Task Failures

Repair and Rerun

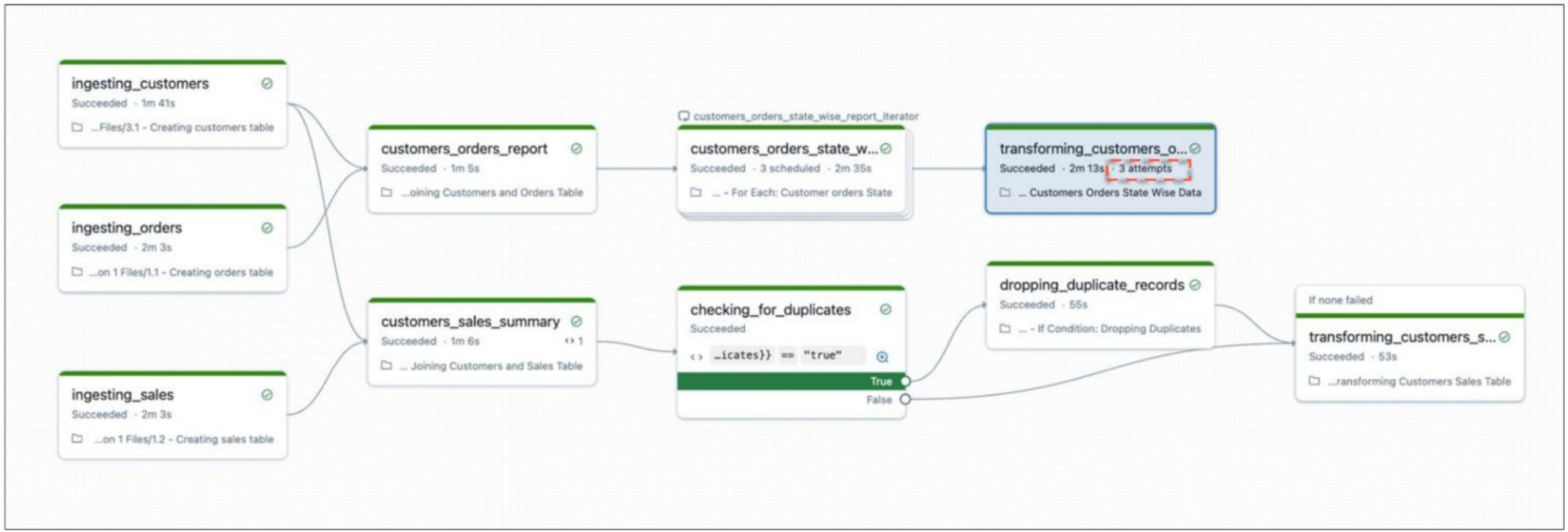


Allowing you to **run only failed tasks**, saving you time and money by only re-running the necessary tasks instead of the entire job

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Handling Task Failures

Repair and Rerun



After re-running task, your final job will look like this.

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Advanced Lakeflow Jobs Features

LECTURE

Monitoring Jobs Performance

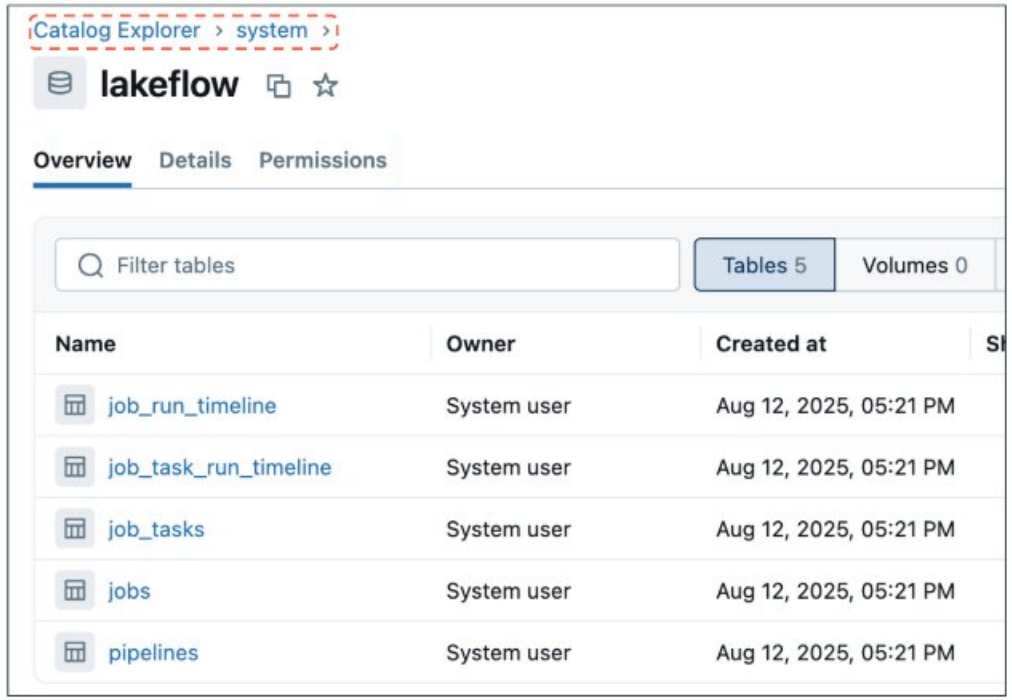


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Monitoring Jobs Performance

System Tables

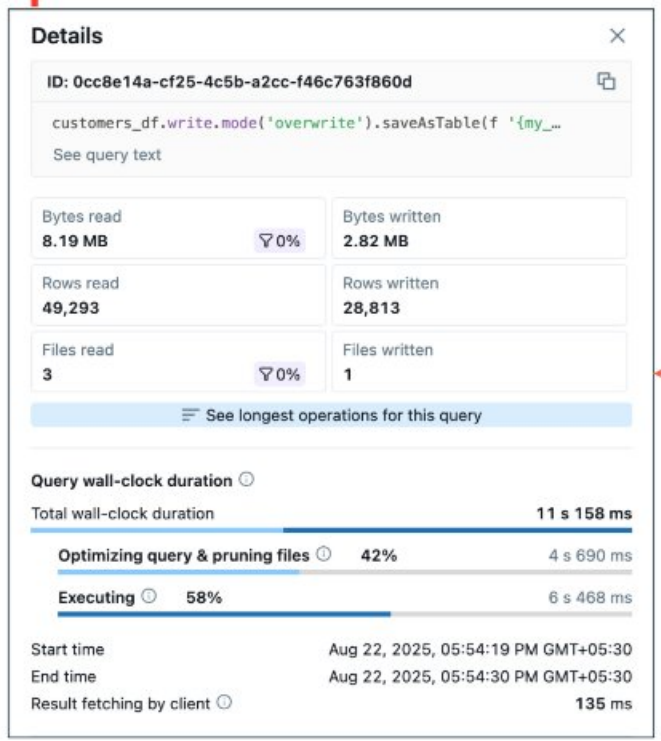
- **system.lakeflow** is a built-in, read-only catalog that logs all job activity across workspaces in the region
- Timeline tables slice long runs hourly using **period_start_time** and **period_end_time**, enabling reliable duration, concurrency, and SLA analytics.
- Key Tables:
  - **jobs** (Job basic info)
  - **job_tasks** (Task basic definitions)
  - **job_run_timeline** (Each job run over time)
  - **job_task_run_timeline** (Each task run over time)
  - **Pipelines** (Pipelines basic info)

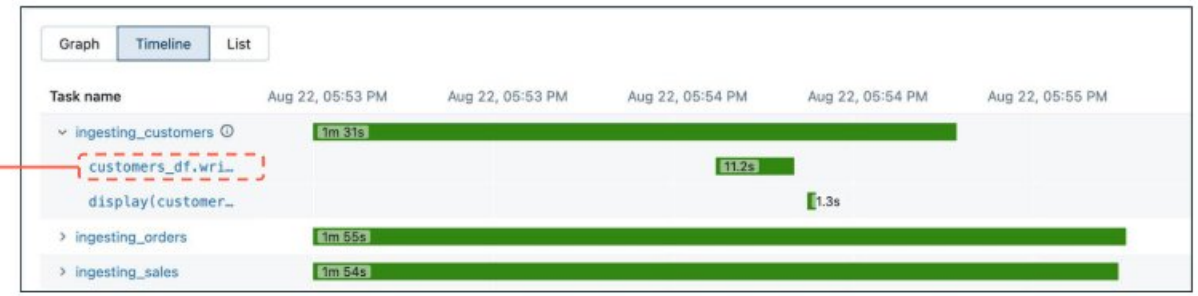


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Monitoring Jobs Performance

Spark UI





- **Timeline** in job run highlights task start/end, duration, and overlap to spot bottlenecks fast.
- Click a task to see status, timestamps, duration, cluster/runtime, logs, and quick I/O.
- Click on **query/code** details for full text, run ID, wall-clock split (optimizing/pruning vs executing), files read/written details, files & partitions and spill details

Act on insights:
High planning time → Improve pruning/partitioning
High execution time → Optimize joins/aggregations (broadcast, skew fixes)

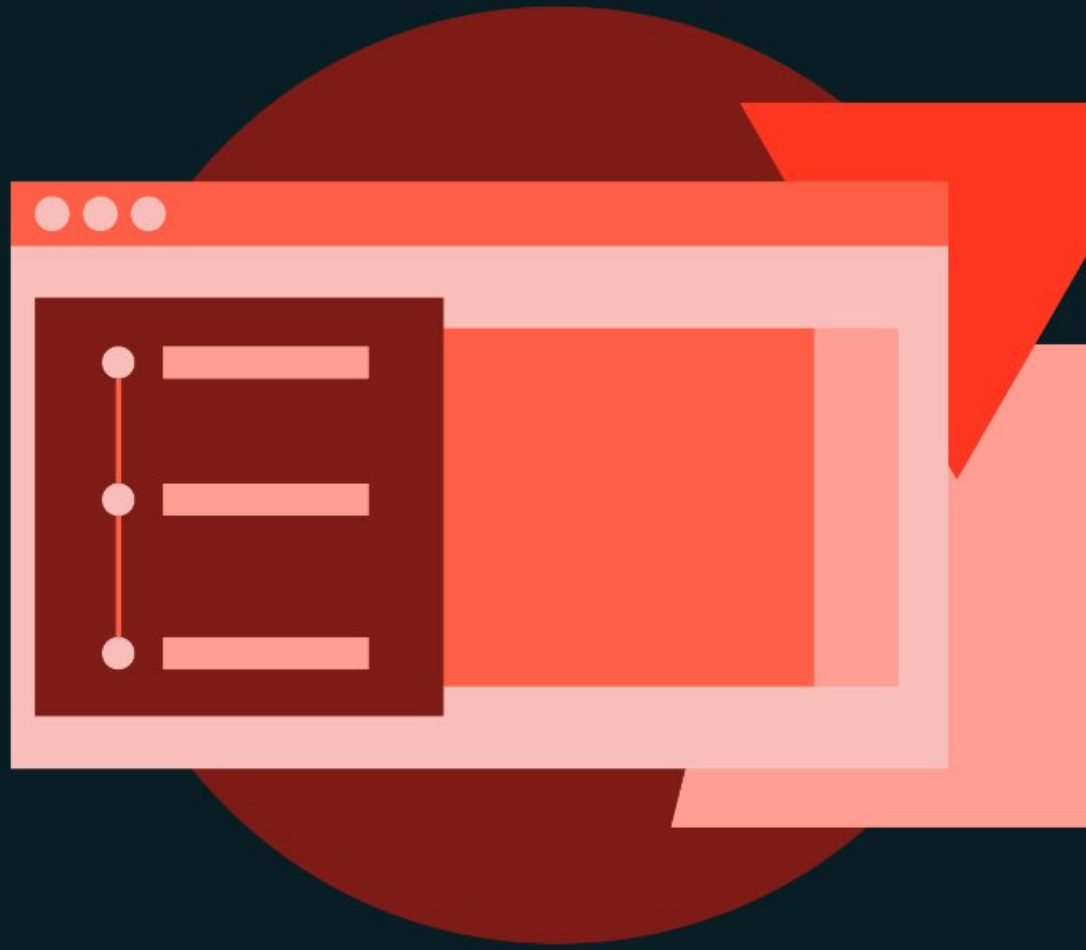
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Advanced Lakeflow Jobs Features

DEMONSTRATION

Monitoring and Repairing Task



Notebook: 6 - Monitoring and Repairing Task

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Advanced Lakeflow Jobs Features

LECTURE

Lakeflow Jobs in Production



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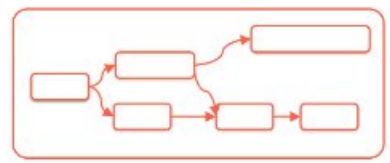
Lakeflow Jobs in Production

Common Best Practices

Compute



Modular Design



Git



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Lakeflow Jobs in Production

Selecting Compute

Interactive Clusters



It is best for performing ad-hoc analysis, data exploration, or development, **but not production**

- Costly for Job runs
- Limited Scalability
- Availability could be an issue because of parallel usage

Job Clusters



It is **cheaper** as they terminate when the job ends, reducing resource usage and costs. But,

- Start-up Latency
- Subject to cloud provider start-up time
- Maintenance Burden

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Lakeflow Jobs in Production

Selecting Compute

Serverless Compute



Simplicity: You no longer need to choose the VM type. Customers do not need dedicated skilled DevOps folks



High efficiency: Photon turned on default



Faster startup: VMs run in Databricks account. We ensure enough nodes are available based on historical data and ML

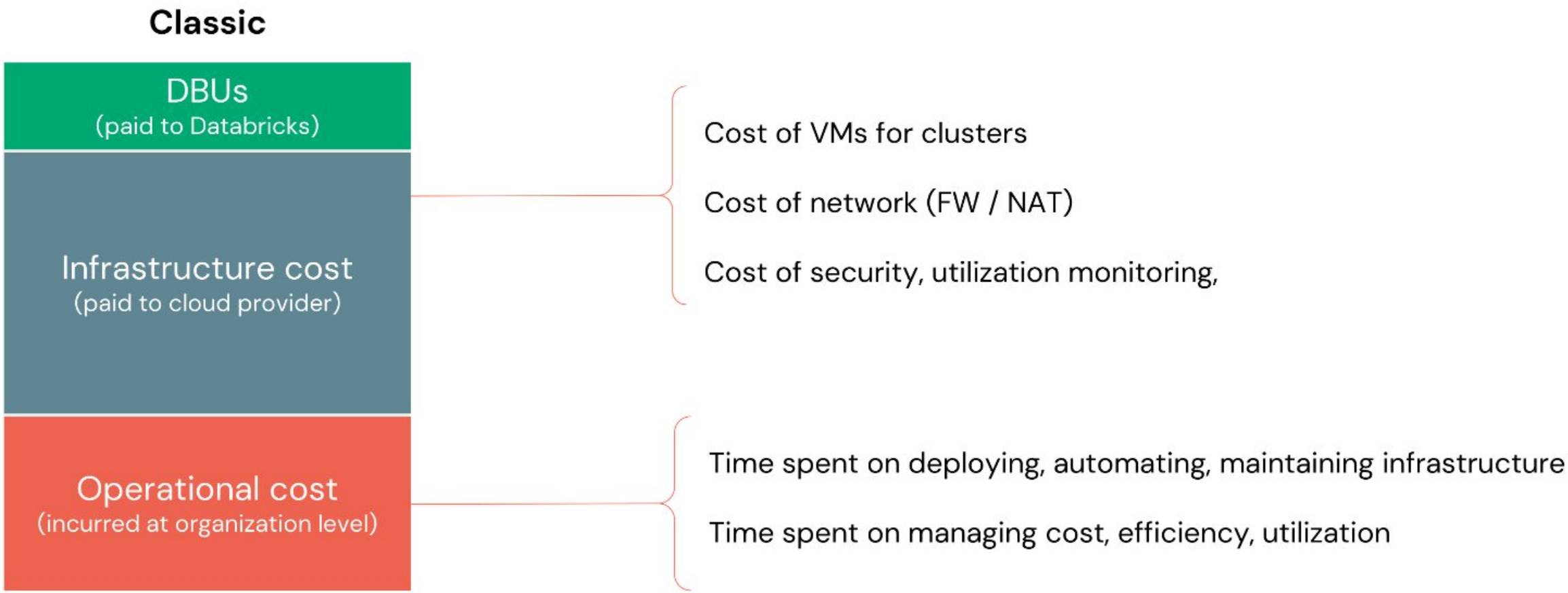


Reliability: Shielded from cloud disruptions

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Lakeflow Jobs in Production

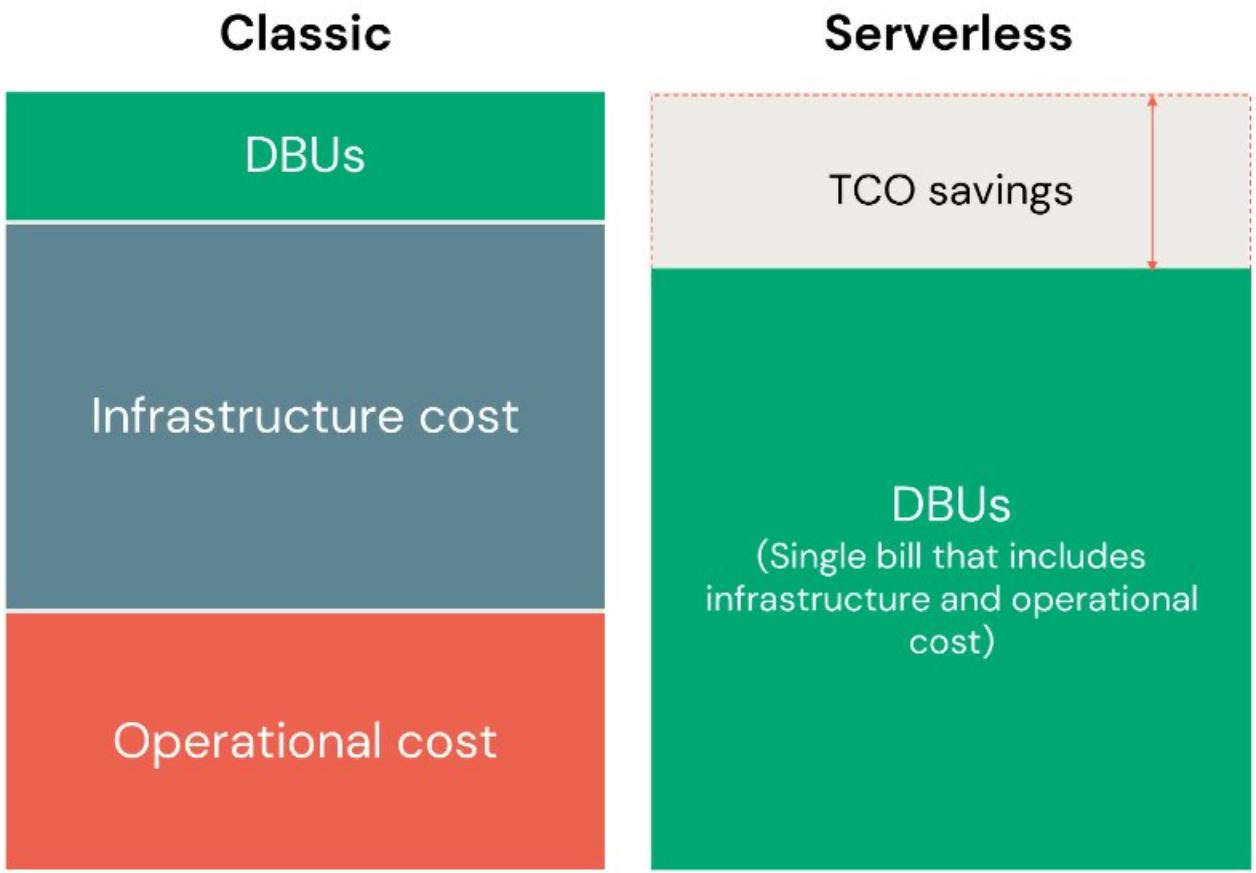
Pricing Structure



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Lakeflow Jobs in Production

Pricing Structure



Value

Fully managed service - operationally simpler, more reliable

Fast clusters, auto-scaling - better user experience, lower cost

Out of the box performance and optimizations - lower overall TCO

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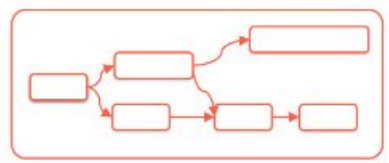
Lakeflow Jobs in Production

Common Best Practices

Compute



Modular Design



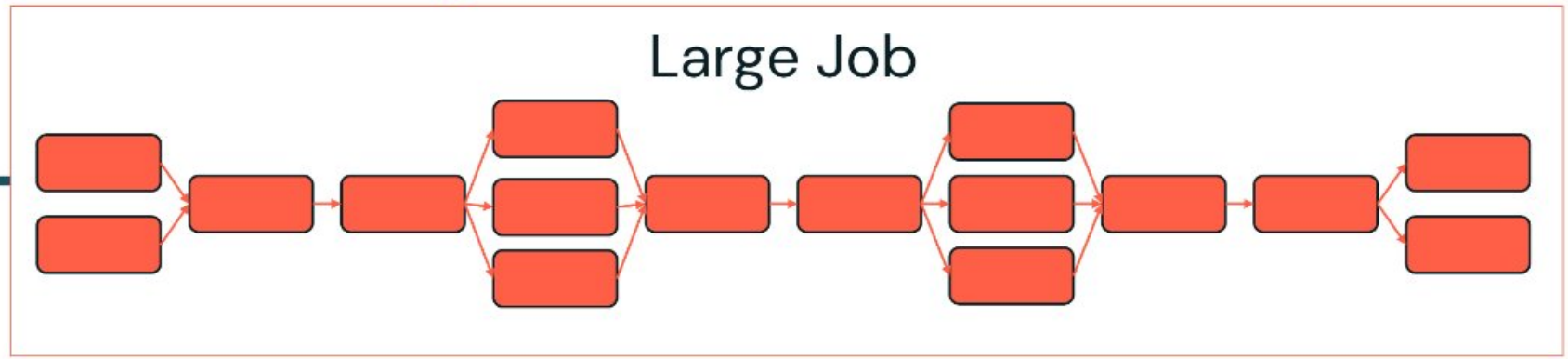
Git



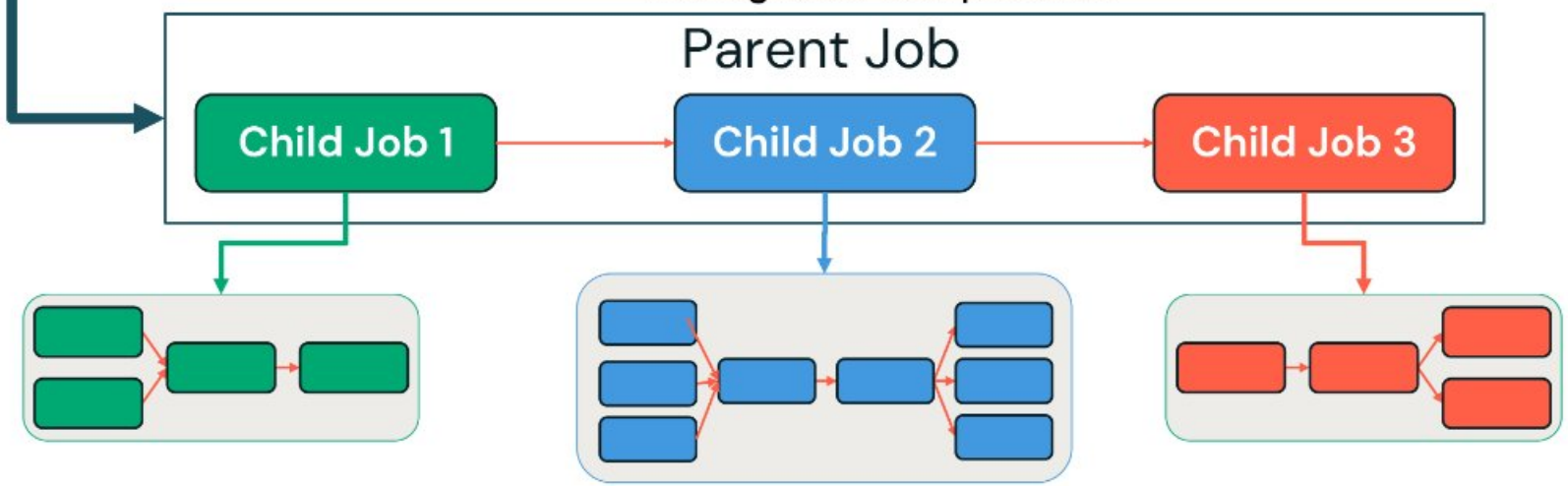
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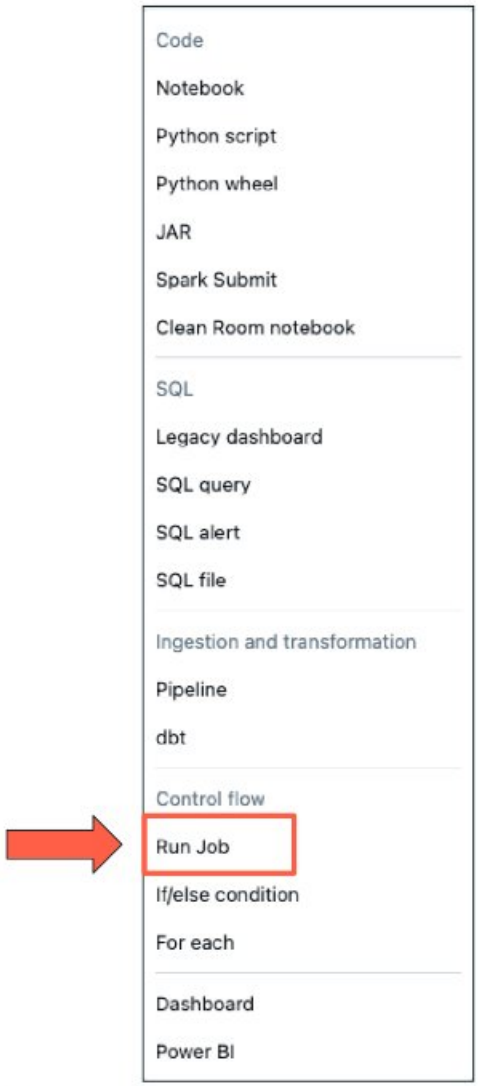
Lakeflow Jobs in Production

Modular Design in Databricks LakeFlow Jobs



Modular orchestration in jobs enables you to break down complex DAGs into smaller, manageable components





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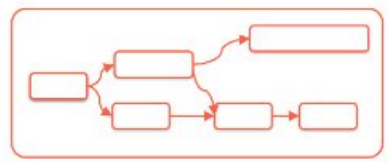
Lakeflow Jobs in Production

Common Best Practices

Compute



Modular Design



Git



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Lakeflow Jobs in Production

Jobs and Git



Jobs supports execution of notebooks from Git

- This **prevents unintentional changes** to your **production** job, for example, when another user makes local edits in a 'Prod' repo or switches branches
- **Simplifying the job definition** process by having a single source of truth
- Makes **deployment** easy for notebooks through CI/CD
- Users can **connect** with **various Git platforms**, as Databricks supports GitHub, GitLab, AWS CodeCommit, and other Git providers.

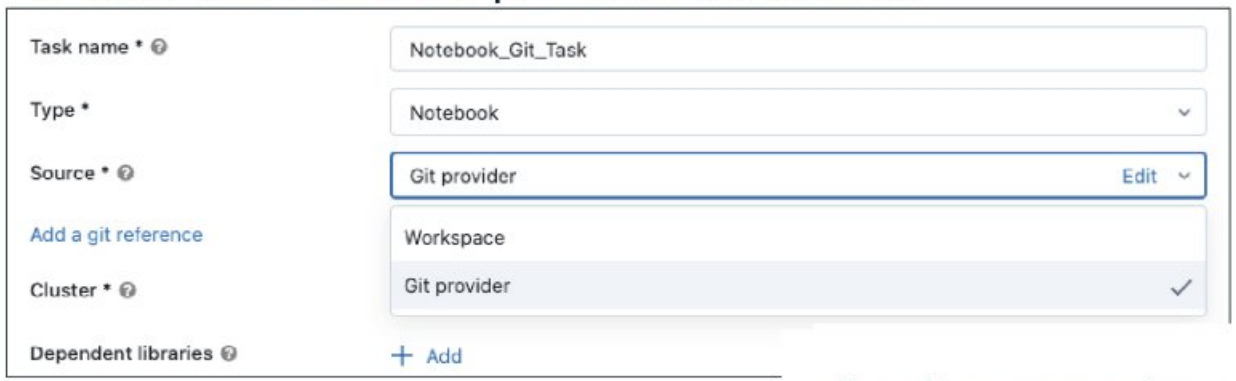
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Lakeflow Jobs in Production

Jobs and Git Configuration Steps

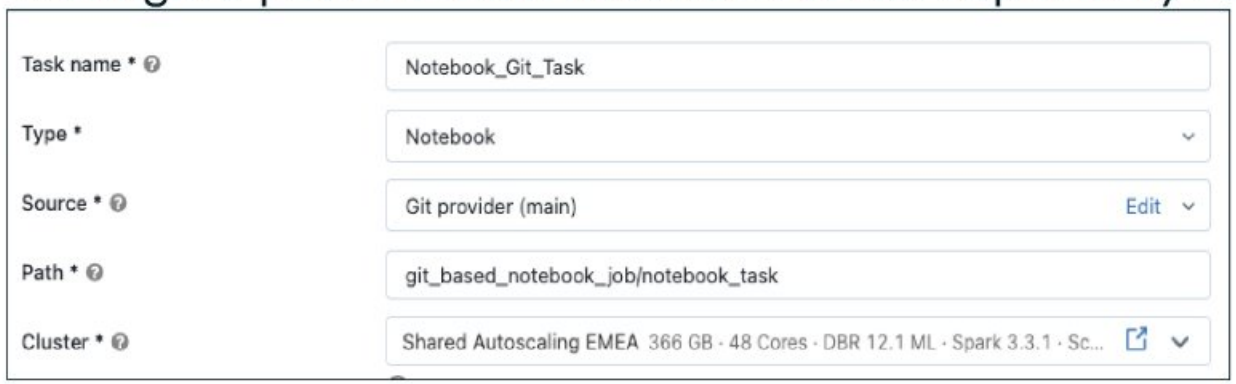
Create task with Git provider as Source

1



Configure path to main notebook under repository root

2



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Advanced Lakeflow Jobs Features

LECTURE

Best Practices



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Best Practices

Compute & Cost Optimization



- Use job or **serverless clusters** in production
- Avoid interactive clusters for production workloads
- **Enable Photon** for faster and cheaper execution
- **Reuse clusters** to reduce startup time and cost

Orchestration & Modularity



- **Break complex pipelines** into modular jobs/run jobs
- Use **multi-task jobs** for parallel and scalable execution
- Apply conditional logic using Run If, If/Else, For Each task
- Limit jobs to **manageable task counts** for maintainability (Limited to 1000 task)

Monitoring & Governance



- **Use service principals** for job ownership and data access
- Configure alerts for **failures, delays, and completions**
- Leverage **Repair & Run** for reducing cost on re-processing
- Parameterize tasks for **reusability and flexibility**

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Summary and Next Steps



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Course Learning Objectives Recap

- Understand the role of Lakeflow Jobs within the Databricks ecosystem as a unified orchestration platform for data, analytics, and AI workloads.
- Design and implement data workloads using Directed Acyclic Graphs (DAGs), demonstrating the relationship between jobs, tasks, and dependencies.
- Configure various job scheduling options including manual, scheduled, file arrival, and continuous triggers to automate workflow execution.
- Implement advanced workflow features such as conditional task execution, run-if dependencies, and repair runs to create robust, fault-tolerant data pipelines.
- Apply best practices for production workloads including selecting appropriate compute options, implementing modular orchestration, and utilizing proper error handling techniques.



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Next Steps

Additional resources for continuing the learning journey.

Data Engineering with Databricks

- Continue your learning through self-paced or instructor-led offerings
- The courses offer hands-on instruction in:
  - Databricks Data Science & Engineering Workspace
  - Databricks SQL
  - Databricks Repos
  - Databricks Task Orchestration
  - Unity Catalog

Data Engineer Associate Certification

- Validate your data and AI skills on Databricks by earning a Databricks credential
- Exam information and exam guide
- The exam covers:
  - Data Intelligence Platform
  - ELT With Spark SQL and Python
  - Incremental Data Processing
  - Production Pipelines
  - Data Governance



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Advanced Lakeflow Jobs Features

LAB EXERCISE

BONUS – Modular Orchestration



This is an optional bonus lab that can be completed after class if you're interested in practicing modular design.

You will most likely not have time to complete this bonus lab during a live class.

Notebook: 7 BONUS LAB - Modular Orchestration

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